

19. Buchheit M. The numbers will love you back in return-I promise. *Int J Sports Physiol Perform*. 2016;11(4):551-4.
20. Buchheit M, Simpson BM. Player tracking technology: Half-full or half-empty glass? *Int J Sports Physiol Perform*. 2017;12(Suppl 2):S235-S41.
21. Buchheit M, Gray A, Morin JB. Assessing stride variables and vertical stiffness with GPS-embedded accelerometers: Preliminary insights for the monitoring of neuromuscular fatigue on the field. *J Sports Sci Med*. 2015;14(4):698-701.
22. Buchheit M, Manouvrier C, Cassirame J, Morin JB. Monitoring locomotor load in soccer: Is metabolic power, powerful? *Int J Sports Med*. 2015;36(14):1149-55.
23. Buchheit M, Allen A, Poon TK, Modonutti M, Gregson W, Di Salvo V. Integrating different tracking systems in football: Multiple camera semi-automatic system, local position measurement and GPS technologies. *J Sports Sci*. 2014;32(20)(20):1844-57.
24. Buchheit M, Al Haddad H, Simpson BM, Palazzi D, Bourdon PC, Di Salvo V, et al. Monitoring accelerations with GPS in football: Time to slow down? *Int J Sports Physiol Perform*. 2014;9(3):442-5.
25. Buchheit M, Hammond K, Bourdon PC, Simpson BM, Garvican-Lewis LA, Schmidt WF, et al. Relative match intensities at high altitude in highly-trained young soccer players (ISA3600). *J Sports Sci Med*. 2015;14(1):98-102.
26. Cardinale M, Varley MC. Wearable training-monitoring technology: Applications, challenges, and opportunities. *Int J Sports Physiol Perform*. 2017;12(Suppl 2):S255-s62.
27. Cerretelli P, Di Prampero PE. Kinetics of respiratory gas exchange and cardiac output at the onset of exercise. *Scand J Respir Dis*. 1971;Suppl (77):35a-g.
28. Cobbaert C, Morales C, van Fessem M, Kemperman H. Precision, accuracy and linearity of radiometer EML 105 whole blood metabolite biosensors. *Ann Clin Biochem*. 1999;36(Pt 6):730-8.
29. Coutts AJ. In the age of technology, Occam's razor still applies. *Int J Sports Physiol Perform*. 2014;9(5):741.
30. Coutts AJ, Rampinini E, Marcora SM, Castagna C, Impellizzeri FM. Heart rate and blood lactate correlates of perceived exertion during small-sided soccer games. *J Sci Med Sport*. 2009;12(1):79-84.
31. Coyle EF, Martin WH, Ehsani AA, Hagberg JM, Bloomfield SA, Sinacore DR, et al. Blood lactate threshold in some well-trained ischemic heart disease patients. *J Appl Physiol*. 1983;54(1):18-23.
32. Coyle EF, Feltner ME, Kautz SA, Hamilton MT, Montain SJ, Baylor AM, et al. Physiological and biomechanical factors associated with elite endurance cycling performance. *Med Sci Sports Exerc*. 1991;23(1):93-107.
33. Dascombe BJ, Reaburn PR, Sirotic AC, Coutts AJ. The reliability of the i-STAT clinical portable analyser. *J Sci Med Sport/Sports Med Aust*. 2007;10(3):135-40.
34. di Prampero PE, Fusi S, Sepulcri L, Morin JB, Belli A, Antonutto G. Sprint running: A new energetic approach. *J Exp Biol*. 2005;208(Pt 14):2809-16.
35. Duhig S, Shield AJ, Opar D, Gabbett TJ, Ferguson C, Williams M. Effect of high-speed running on hamstring strain injury risk. *Br J Sports Med*. 2016;50(24):1536-40.
36. Fell JW, Rayfield JM, Gulbin JP, Gaffney PT. Evaluation of the Accusport Lactate Analyser. *Int J Sports Med*. 1998;19(3):199-204.
37. Foster C, Hector LL, Welsh R, Schrager M, Green MA, Snyder AC. Effects of specific versus cross-training on running performance. *Eur J Appl Physiol Occup Physiol*. 1995;70(4):367-72.
38. Hader K, Mendez-Villanueva A, Palazzi D, Ahmaidi S, Buchheit M. Metabolic power requirement of change of direction speed in young soccer players: Not all is what it seems. *PLoS One*. 2016;11(3):e0149839.
39. Hagglund M, Walden M, Magnusson H, Kristenson K, Bengtsson H, Ekstrand J. Injuries affect team performance negatively in professional football: An 11-year follow-up of the UEFA Champions League injury study. *Br J Sports Med*. 2013;47(12):738-42.
40. Harris NK, Cronin J, Taylor KL, Boris J, Sheppard J. Understanding position transducer technology for strength and conditioning practitioners. *Strength Cond J*. 2010;32(4):66-79.

41. Haugen T, Buchheit M. Sprint running performance monitoring: Methodological and practical considerations. *Sports Med.* 2016;46(5):641-56.
42. Highton J, Mullen T, Norris J, Oxendale C, Twist C. Energy expenditure derived from micro-technology is not suitable for assessing internal load in collision-based activities. *Int J Sports Physiol Perform.* 2017;12(2):264-7.
43. Hill-Haas S, Coutts A, Rowsell G, Dawson B. Variability of acute physiological responses and performance profiles of youth soccer players in small-sided games. *J Sci Med Sport.* 2008;11(5):487-90.
44. Impellizzeri FM, Rampinini E, Coutts AJ, Sassi A, Marcora SM. Use of RPE-based training load in soccer. *Med Sci Sports Exerc.* 2004;36(6):1042-7.
45. Laursen PB, Jenkins DG. The scientific basis for high-intensity interval training: Optimising training programmes and maximising performance in highly trained endurance athletes. *Sports Med.* 2002;32(1):53-73.
46. Malone JJ, Lovell R, Varley MC, Coutts AJ. Unpacking the black box: Applications and considerations for using GPS devices in sport. *Int J Sports Physiol Perform.* 2017;12(Suppl 2):S218-S26.
47. Manzi V, Bovenzi A, Franco Impellizzeri M, Carminati I, Castagna C. Individual training-load and aerobic-fitness variables in premiership soccer players during the precompetitive season. *J Strength Cond Res.* 2013;27(3):631-6.
48. Marsland F, Mackintosh C, Anson J, Lyons K, Waddington G, Chapman DW. Using micro-sensor data to quantify macro kinematics of classical cross-country skiing during on-snow training. *Sports Biomech.* 2015;14(4):435-47.
49. Marsland F, Mackintosh C, Holmberg HC, Anson J, Waddington G, Lyons K, et al. Full course macro-kinematic analysis of a 10 km classical cross-country skiing competition. *PLoS One.* 2017;12(8):e0182262.
50. Mendez-Villanueva A, Buchheit M, Simpson BM, Bourdon PC. Match play intensity distribution in youth soccer. *Int J Sport Med.* 2013;34(2):101-10.
51. Michael JS, Rooney KB, Smith RM. The dynamics of elite paddling on a kayak simulator. *J Sports Sci.* 2012;30(7):661-8.
52. Midgley AW, McNaughton LR, Carroll S. Reproducibility of time at or near VO₂max during intermittent treadmill running. *Int J Sports Med.* 2007;28(1):40-7.
53. Nielsen OB, de Paoli F, Overgaard K. Protective effects of lactic acid on force production in rat skeletal muscle. *J Physiol.* 2001;536(Pt 1):161-6.
54. Osgnach C, Poser S, Bernardini R, Rinaldo R, di Prampero PE. Energy cost and metabolic power in elite soccer: A new match analysis approach. *Med Sci Sports Exerc.* 2010;42(1):170-8.
55. Quod MJ, Martin DT, Martin JC, Laursen PB. The power profile predicts road cycling MMP. *Int J Sports Med.* 2010;31(6):397-401.
56. Ritchie D, Hopkins WG, Buchheit M, Cordy J, Bartlett JD. Quantification of training load during return to play after upper- and lower-body injury in Australian Rules Football. *Int J Sports Physiol Perform.* 2017;12(5):634-41.
57. Robergs RA, Ghiasvand F, Parker D. Biochemistry of exercise-induced metabolic acidosis. *Am J Physiol Regul Integr Comp Physiol.* 2004;287(3):R502-16.
58. Scott TJ, Black CR, Quinn J, Coutts AJ. Validity and reliability of the session-RPE method for quantifying training in Australian football: A comparison of the CR10 and CR100 scales. *J Strength Cond Res.* 2013;27(1):270-6.
59. Scott TJ, Thornton HR, Scott MTU, Dascombe BJ, Duthie GM. Differences between relative and absolute speed and metabolic thresholds in Rugby League. *Int J Sports Physiol Perform.* 2017:1-21.
60. Seiler S. What is best practice for training intensity and duration distribution in endurance athletes? *Int J Sports Physiol Perform.* 2010;5(3):276-91.
61. Seiler S, Hetlelid KJ. The impact of rest duration on work intensity and RPE during interval training. *Med Sci Sports Exerc.* 2005;37(9):1601-7.
62. Smith TB, Hopkins WG. Measures of rowing performance. *Sports Med.* 2012;42(4):343-58.
63. Stevens TG, de Ruiter CJ, van Maurik D, van Lierop CJ, Savelsbergh GJ, Beek PJ. Measured and estimated energy cost of constant and shuttle running in soccer players. *Med Sci Sports Exerc.* 2015;47(6):1219-24.

64. van Someren KA, Howatson G, Nunan D, Thatcher R, Shave R. Comparison of the Lactate Pro and Analox GM7 blood lactate analysers. *Int J Sports Med*. 2005;26(8):657-61.

Chapter 9

1. Achten J, Jeukendrup AE. Heart rate monitoring: Applications and limitations. *Sports Med*. 2003;33(7):517-38.
2. Akenhead R, Nassis GP. Training load and player monitoring in high-level football: Current practice and perceptions. *Int J Sports Physiol Perform*. 2016;11(5):587-93.
3. Akubat I, Patel E, Barrett S, Abt G. Methods of monitoring the training and match load and their relationship to changes in fitness in professional youth soccer players. *J Sports Sci*. 2012;30(14):1473-80.
4. Al Haddad H, Parouty J, Buchheit M. Effect of daily cold water immersion on heart rate variability and subjective ratings of well-being in highly trained swimmers. *Int J Sports Physiol Perform*. 2012;7(1):33-8.
5. Al Haddad H, Laursen PB, Chollet D, Ahmaidi S, Buchheit M. Reliability of resting and postexercise heart rate measures. *Int J Sports Med*. 2011;32(8):598-605.
6. Aughey RJ. Applications of GPS technologies to field sports. *Int J Sports Physiol Perform*. 2011;6(3):295-310.
7. Barnes KR, Hopkins WG, McGuigan MR, Kilding AE. Effects of different uphill interval-training programs on running economy and performance. *Int J Sports Physiol Perform*. 2013;8(6):639-47.
8. Barrett S, Midgley A, Lovell R. PlayerLoad: Reliability, convergent validity, and influence of unit position during treadmill running. *Int J Sports Physiol Perform*. 2014;9(6):945-52.
9. Bartlett JD, O'Connor F, Pitchford N, Torres-Ronda L, Robertson SJ. Relationships between internal and external training load in team-sport athletes: Evidence for an individualized approach. *Int J Sports Physiol Perform*. 2017;12(2):230-4.
10. Batterham AM, Hopkins WG. Making meaningful inferences about magnitudes. *Int J Sports Physiol Perform*. 2006;1(1):50-7.
11. Beneke R, Leithauser RM, Ochentel O. Blood lactate diagnostics in exercise testing and training. *Int J Sports Physiol Perform*. 2011;6(1):8-24.
12. Berglund B, Safstrom H. Psychological monitoring and modulation of training load of world-class canoeists. *Med Sci Sports Exerc*. 1994;26(8):1036-40.
13. Billman GE, Hoskins RS. Time-series analysis of heart rate variability during submaximal exercise. Evidence for reduced cardiac vagal tone in animals susceptible to ventricular fibrillation. *Circulation*. 1989;80(1):146-57.
14. Borresen J, Lambert MI. Changes in heart rate recovery in response to acute changes in training load. *Eur J Appl Physiol*. 2007;101(4):503-11.
15. Borresen J, Lambert MI. Autonomic control of heart rate during and after exercise: Measurements and implications for monitoring training status. *Sports Med*. 2008;38(8):633-46.
16. Borresen J, Lambert MI. The quantification of training load, the training response and the effect on performance. *Sports Med*. 2009;39(9):779-95.
17. Bosquet L, Gamelin FX, Berthoin S. Is aerobic endurance a determinant of cardiac autonomic regulation? *Eur J Appl Physiol*. 2007;100(3):363-9.
18. Bosquet L, Gamelin FX, Berthoin S. Reliability of postexercise heart rate recovery. *Int J Sports Med*. 2008;29(3):238-43.
19. Bosquet L, Merkari S, Arvisais D, Aubert AE. Is heart rate a convenient tool to monitor overreaching? A systematic review of the literature. *Br J Sports Med*. 2008;42(9):709-14.
20. Bouchard C, Rankinen T. Individual differences in response to regular physical activity. *Med Sci Sports Exerc*. 2001;33:S446-51.
21. Brandenberger G, Buchheit M, Ehrhart J, Simon C, Piquard F. Is slow wave sleep an appropriate recording condition for heart rate variability analysis? *Auton Neurosci*. 2005;121(1-2):81-6.
22. Brenner IK, Thomas S, Shephard RJ. Autonomic regulation of the circulation during exercise and heat exposure. Inferences from heart rate variability. *Sports Med*. 1998;26(2):85-99.
23. Brink MS, Visscher C, Coutts AJ, Lemmink KA. Changes in perceived stress and recovery in overreached young elite soccer players. *Scand J Med Sci Sports*. 2010:[Epub ahead of print] Oct 7.
24. Buchheit M. Monitoring training status with HR measures: Do all roads lead to Rome? *Front Physiol*. 2014;27(5):73.

25. Buchheit M. The numbers will love you back in return—I promise. *Int J Sports Physiol Perform*. 2016;11(4):551-4.
26. Buchheit M. Applying the acute:chronic workload ratio in elite football: Worth the effort? *Br J Sports Med*. 2017;51(18):1325-7.
27. Buchheit M. Want to see my report, coach? *Aspetar Journal*. 2017(6):34-43. <http://www.aspetar.com/journal/upload/PDF/2017216161135.pdf>.
28. Buchheit M. Trivial effects are clearly important. *Sport Perform Sci Rep*. Jan 5, 2018;1.
29. Buchheit M. Magnitudes matter more than beetroot juice. *Sport Perform Sci Rep*. Jan 15, 2018;1.
30. Buchheit M, Gindre C. Cardiac parasympathetic regulation: Respective associations with cardio-respiratory fitness and training load. *Am J Physiol Heart Circ Physiol*. 2006;291(1):H451-H8.
31. Buchheit M, Laursen PB. High-intensity interval training, solutions to the programming puzzle. Part II: Anaerobic energy, neuromuscular load and practical applications. *Sports Med*. 2013;43(10):927-54.
32. Buchheit M, Rabbani A. 30-15 Intermittent Fitness Test vs. Yo-Yo Intermittent Recovery Test Level 1: Relationship and sensitivity to training. *Int J Sports Physiol Perform*. 2014;9(3):522-4.
33. Buchheit M, Simpson BM. Player tracking technology: Half-full or half-empty glass? *Int J Sports Physiol Perform*. 2017;12(Suppl 2):S235–S41.
34. Buchheit M, Gray A, Morin JB. Assessing stride variables and vertical stiffness with GPS-embedded accelerometers: Preliminary insights for the monitoring of neuromuscular fatigue on the field. *J Sports Sci Med*. 2015;14(4):698-701.
35. Buchheit M, Cholley Y, Lambert P. Psychometric and physiological responses to a preseason competitive camp in the heat with a 6-hour time difference in elite soccer players. *Int J Sports Physiol Perform*. 2016;11(2):176-81.
36. Buchheit M, Papelier Y, Laursen PB, Ahmaidi S. Noninvasive assessment of cardiac parasympathetic function: Post-exercise heart rate recovery or heart rate variability? *Am J Physiol Heart Circ Physiol*. 2007;293(1):H8-H10.
37. Buchheit M, Laursen PB, Al Haddad H, Ahmaidi S. Exercise-induced plasma volume expansion and post-exercise parasympathetic reactivation. *Eur J Appl Physiol*. 2009;105(3):471-81.
38. Buchheit M, Mendez-Villanueva A, Quod M, Bourdon P. Determinants of the variability of heart rate measures during a competitive period in young soccer players. *Eur J Appl Physiol*. 2010;109:869-78.
39. Buchheit M, Lacomme M, Cholley Y, Simpson BM. Neuromuscular responses to conditioned soccer sessions assessed via GPS-embedded accelerometers: Insights into tactical periodization. *Int J Sports Physiol Perform*. Sep 2017;5:1-21.
40. Buchheit M, Simon C, Piquard F, Ehrhart J, Brandenberger G. Effect of increased training load on vagal-related indexes of heart rate variability: A novel sleep approach. *Am J Physiol Heart Circ Physiol*. 2004;287:H2813-H8.
41. Buchheit M, Voss SC, Nybo L, Mohr M, Racinais S. Physiological and performance adaptations to an in-season soccer camp in the heat: Associations with heart rate and heart rate variability. *Scand J Med Sci Sports*. 2011;21(6):e477-85.
42. Buchheit M, Al Haddad H, Mendez-Villanueva A, Quod MJ, Bourdon PC. Effect of maturation on hemodynamic and autonomic control recovery following maximal running exercise in highly trained young soccer players. *Front Physiol*. 2011;2:69.
43. Buchheit M, Simpson MB, Al Haddad H, Bourdon PC, Mendez-Villanueva A. Monitoring changes in physical performance with heart rate measures in young soccer players. *Eur J Appl Physiol*. 2012;112(2):711-23.
44. Buchheit M, Simon C, Charloux A, Doutreleau S, Piquard F, Brandenberger G. Relationship between very high physical activity energy expenditure, heart rate variability and self-estimate of health status in middle-aged individuals. *Int J Sports Med*. 2006;27(9):697-701.
45. Buchheit M, Millet GP, Parisy A, Pourchez S, Laursen PB, Ahmaidi S. Supramaximal training and post-exercise parasympathetic reactivation in adolescents. *Med Sci Sports Exerc*. 2008;40(2):362-71.
46. Buchheit M, Chivot A, Parouty J, Mercier D, Al Haddad H, Laursen PB, et al. Monitoring endurance running performance using cardiac parasympathetic function. *Eur J Appl Physiol*. 2010;108:1153-67.

47. Buchheit M, Racinais S, Bilsborough JC, Bourdon PC, Voss SC, Hocking J, et al. Monitoring fitness, fatigue and running performance during a pre-season training camp in elite football players. *J Sci Med Sport*. 2013;16(6):550-5.
48. Buchheit M, Simpson BM, Schmidt W, Aughey RJ, Soria R, Hunt R, et al. Predicting sickness during a 2-week soccer camp at 3600 m (ISA3600). *Br J Sports Med*. 2013;47(Suppl 1):i124-i17.
49. Buchheit M, Simpson BM, Garvican-Lewis LA, Hammond K, Kley M, Schmidt WF, et al. Wellness, fatigue and physical performance acclimatisation to a 2-week soccer camp at 3600 m (ISA3600). *Br J Sports Med*. 2013;(47 Suppl):i100-i6.
50. Burgess DJ. The research doesn't always apply: Practical solutions to evidence-based training-load monitoring in elite team sports. *Int J Sports Physiol Perform*. 2017;12 (Suppl 2):S2136-s41.
51. Burton AR, Rahman K, Kadota Y, Lloyd A, Vollmer-Conna U. Reduced heart rate variability predicts poor sleep quality in a case-control study of chronic fatigue syndrome. *Exp Brain Res*. 2010;204(1):71-8.
52. Carey DL, Blanch P, Ong KL, Crossley KM, Crow J, Morris ME. Training loads and injury risk in Australian football-differing acute:chronic workload ratios influence match injury risk. *Br J Sports Med*. 2017;51(16):1215-20.
53. Carling C. Interpreting physical performance in professional soccer match-play: Should we be more pragmatic in our approach? *Sports Med*. 2013;43(8):655-63.
54. Cassirame J, Stuckey MI, Sheppard F, Tordi N. Accuracy of the Minicardio system for heart rate variability analysis compared to ECG. *J Sports Med Phys Fitness*. 2013;53(3):248-54.
55. Castagna C, Impellizzeri FM, Chaouachi A, Bordon C, Manzi V. Effect of training intensity distribution on aerobic fitness variables in elite soccer players: A case study. *J Strength Cond Res*. 2011;25(1):66-71.
56. Cerretelli P, Di Prampero PE. Kinetics of respiratory gas exchange and cardiac output at the onset of exercise. *Scand J Respir Dis*. 1971;(Suppl 77):35a-g.
57. Chen JL, Yeh DP, Lee JP, Chen CY, Huang CY, Lee SD, et al. Parasympathetic nervous activity mirrors recovery status in weightlifting performance after training. *J Strength Cond Res*. 2011;25(6):1546-52.
58. Cormack SJ, Newton RU, McGuigan MR, Cormie P. Neuromuscular and endocrine responses of elite players during an Australian rules football season. *Int J Sports Physiol Perform*. 2008;3(4):439-53.
59. Daanen HA, Lamberts RP, Kallen VL, Jin A, Van Meeteren NL. A systematic review on heart-rate recovery to monitor changes in training status in athletes. *Int J Sports Physiol Perform*. 2012;7(3):251-60.
60. Dellaserra CL, Gao Y, Ransdell L. Use of integrated technology in team sports: A review of opportunities, challenges, and future directions for athletes. *J Strength Cond Res*. 2014;28(2):556-73.
61. Drew MK, Finch CF. The relationship between training load and injury, illness and soreness: A systematic and literature review. *Sports Med*. 2016;46(6):861-83.
62. Dupuy O, Mekary S, Berryman N, Bherer L, Audiffren M, Bosquet L. Reliability of heart rate measures used to assess post-exercise parasympathetic reactivation. *Clin Physiol Funct Imaging*. 2012;32(4):296-304.
63. Ekkekakis P. *The Measurement of Affect, Mood, and Emotion: A Guide for Health-Behavioral Research*. New York: Cambridge University Press; 2013.
64. Esco MR, Flatt AA. Ultra-short-term heart rate variability indexes at rest and post-exercise in athletes: Evaluating the agreement with accepted recommendations. *J Sports Sci Med*. 2014;13(3):535-41.
65. Fanchini M, Rampinini E, Riggio M, Coutts A, Pecci C, McCall A. Despite association, the acute:chronic work load ratio does not predict non-contact injury in elite footballers. *Science and Medicine in Football*. 2018: In press.
66. Flatt AA, Esco MR. Validity of the ithleteTM smart phone application for determining ultra-short-term heart rate variability. *Journal of Human Kinetics*. 2013;39:85-92.
67. Fowler PM, Murray A, Farooq A, Lumley N, Taylor L. Subjective and objective responses to two Rugby 7's World Series competitions. *Journal of Strength and Conditioning Research*. [Ahead of print.]

68. Gamelin FX, Berthoin S, Bosquet L. Validity of the polar S810 heart rate monitor to measure R-R intervals at rest. *Med Sci Sports Exerc.* 2006;38(5):887-93.
69. Garet M, Tournaire N, Roche F, Laurent R, Lacour JR, Barthelemy JC, et al. Individual interdependence between nocturnal ANS activity and performance in swimmers. *Med Sci Sports Exerc.* 2004;36(12):2112-8.
70. Garvican LA, Hammond K, Varley M, Gore CJ, Billaut F, Aughey RJ. Lower running performance and exacerbated fatigue in soccer played at 1600 m. *Int J Sports Physiol Perform.* May 22, 2013. [Epub ahead of print].
71. Green HJ, Thomson JA, Ball ME, Hughson RL, Houston ME, Sharratt MT. Alterations in blood volume following short-term supramaximal exercise. *J Appl Physiol.* 1984;56(1):145-9.
72. Halson SL. Monitoring training load to understand fatigue in athletes. *Sports Med.* 2014;44(Suppl 2):S139-47.
73. Hamilton DK. Explosive performance in young soccer players during several games in succession: a tournament scenario. *J Aust Strength Cond.* 2009;17(2):3-10.
74. Hamilton RM, McKechnie PS, Macfarlane PW. Can cardiac vagal tone be estimated from the 10-second ECG? *Int J Cardiol.* 2004;95(1):109-15.
75. Hampson R, Jowett S. Effects of coach leadership and coach-athlete relationship on collective efficacy. *Scand J Med Sci Sports.* 2014;24(2):454-60.
76. Hautala AJ, Kiviniemi AM, Tulppo MP. Individual responses to aerobic exercise: The role of the autonomic nervous system. *Neurosci Biobehav Rev.* 2009;33(2):107-15.
77. Hautala AJ, Tulppo MP, Makikallio TH, Laukkanen R, Nissila S, Huikuri HV. Changes in cardiac autonomic regulation after prolonged maximal exercise. *Clin Physiol.* 2001;21(2):238-45.
78. Hautala AJ, Kiviniemi AM, Makikallio TH, Kinnunen H, Nissila S, Huikuri HV, et al. Individual differences in the responses to endurance and resistance training. *Eur J Appl Physiol.* 2006;96(5):535-42.
79. Hedelin R, Bjerle P, Henriksson-Larsen K. Heart rate variability in athletes: Relationship with central and peripheral performance. *Med Sci Sports Exerc.* 2001;33(8):1394-8.
80. Hedman AE, Hartikainen JE, Tahvanainen KU, Hakumaki MO. The high frequency component of heart rate variability reflects cardiac parasympathetic modulation rather than parasympathetic "tone." *Acta Physiol Scand.* 1995;155(3):267-73.
81. Heisterberg MF, Fahrenkrug J, Andersen JL. Multiple blood samples in elite soccer players. Is it worthwhile? *J Sports Sci.* 2014;32(13):1324-7.
82. Heisterberg MF, Fahrenkrug J, Krstrup P, Storskov A, Kjaer M, Andersen JL. Extensive monitoring through multiple blood samples in professional soccer players. *J Strength Cond Res.* 2013;27(5):1260-71.
83. Hooper SL, Mackinnon LT. Monitoring overtraining in athletes. Recommendations. *Sports Med.* 1995;20(5):321-7.
84. Hopkins W. Spreadsheets for analysis of controlled trials, with adjustment for a subject characteristic. *Sportscience.* 2006;10:46-50. Accessed July 4, 2009. Available at <http://sportscci.org/6/wghcontrial.htm>.
85. Hopkins WG. Measures of reliability in sports medicine and science. *Sports Med.* 2000;30(1):1-15.
86. Hopkins WG. Precision of the estimate of a subject's true value (Excel spreadsheet) (2000). In Internet Society for Sport Science: www.sportsci.org/resource/stats/xprecisionsubject.xls2000 (April 25, 2014).
87. Hopkins WG. Statistical vs clinical or practical significance. *Sportscience* 2002;6. http://www.sportsci.org/jour/0201/Statistical_vs_clinical.ppt.
88. Hopkins WG. How to interpret changes in an athletic performance test. *Sportscience.* 2004; 8:1-7.
89. Hopkins WG, Hawley JA, Burke LM. Design and analysis of research on sport performance enhancement. *Med Sci Sports Exerc.* 1999;31(3):472-85.
90. Hopkins WG, Marshall SW, Batterham AM, Hanin J. Progressive statistics for studies in sports medicine and exercise science. *Med Sci Sports Exerc.* 2009;41(1):3-13.
91. Hug B, Heyer L, Naef N, Buchheit M, Werhlin JP, Millet GP. Tapering for marathon and cardiac autonomic function. *Jul* 2014;35(8):676-83.
92. Hulin BT, Gabbett TJ, Lawson DW, Caputi P, Sampson JA. The acute:chronic workload ratio

- predicts injury: High chronic workload may decrease injury risk in elite rugby league players. *Br J Sports Med*. 2016;50(4):231-6.
93. Iellamo F, Legramante JM, Pigozzi F, Spataro A, Norbiato G, Lucini D, et al. Conversion from vagal to sympathetic predominance with strenuous training in high-performance world class athletes. *Circulation*. 2002;105(23):2719-24.
 94. Impellizzeri FM, Rampinini E, Coutts AJ, Sassi A, Marcora SM. Use of RPE-based training load in soccer. *Med Sci Sports Exerc*. 2004;36(6):1042-7.
 95. Iwasaki K, Zhang R, Zuckerman JH, Levine BD. Dose-response relationship of the cardiovascular adaptation to endurance training in healthy adults: How much training for what benefit? *J Appl Physiol*. 2003;95(4):1575-83.
 96. Kellmann M, Kallus K. *Recovery-Stress Questionnaire for Athletes: User manual*. Champaign, IL: Human Kinetics; 2001.
 97. Kiely J. Periodization paradigms in the 21st century: Evidence-led or tradition-driven? *Int J Sports Physiol Perform*. 2012;7(3):242-50.
 98. Kiviniemi AM, Hautala AJ, Kinnunen H, Tulppo MP. Endurance training guided individually by daily heart rate variability measurements. *Eur J Appl Physiol*. 2007;101(6):743-51.
 99. Kiviniemi AM, Hautala AJ, Seppanen T, Makikallio TH, Huikuri HV, Tulppo MP. Saturation of high-frequency oscillations of R-R intervals in healthy subjects and patients after acute myocardial infarction during ambulatory conditions. *Am J Physiol Heart Circ Physiol*. 2004;287(5):H1921-H7.
 100. Kiviniemi AM, Hautala AJ, Kinnunen H, Nissila J, Virtanen P, Karjalainen J, et al. Daily Exercise prescription based on heart rate variability among men and women. *Med Sci Sports Exerc*. Jul 2010;42(7):1355-63.
 101. Kukiela M, Seals DR, Billman GE. Cardiac vagal modulation of heart rate during prolonged submaximal exercise in animals with healed myocardial infarctions: Effects of training. *Am J Physiol Heart Circ Physiol*. 2006;290(4):H1680-5.
 102. Lacombe M, Simpson BM, Buchheit M. Monitoring training status with player-tracking technology. Still on the way to Rome. *Aspetar Journal*. 2018: In Press.
 103. Ladell WS. Assessment of group acclimatization to heat and humidity. *J Physiol*. 1951;115(3):296-312.
 104. Lamberts RP, Lemmink KA, Durandt JJ, Lambert MI. Variation in heart rate during submaximal exercise: Implications for monitoring training. *J Strength Cond Res*. 2004;18(3):641-5.
 105. Lamberts RP, Swart J, Noakes TD, Lambert MI. Changes in heart rate recovery after high-intensity training in well-trained cyclists. *Eur J Appl Physiol*. 2009;105(5):705-13.
 106. Lamberts RP, Maskell S, Borresen J, Lambert MI. Adapting workload improves the measurement of heart rate recovery. *Int J Sports Med*. 2011;32(9):698-702.
 107. Laursen PB. Training for intense exercise performance: High-intensity or high-volume training? *Scand J Med Sci Sports*. 2010;20(Suppl 2):1-10.
 108. Le Meur Y, Buchheit M, Aubry A, Coutts AJ, Hausswirth C. Assessing overreaching with heart-rate recovery: What is the minimal exercise intensity required? *Int J Sports Physiol Perform*. 2017;12(4):569-73.
 109. Le Meur Y, Hausswirth C, Natta F, Couturier A, Bignet F, Vidal PP. A multidisciplinary approach to overreaching detection in endurance trained athletes. *J Appl Physiol* (1985). 2013;114(3):411-20.
 110. Le Meur Y, Pichon A, Schaal K, Schmitt L, Louis J, Gueneron J, et al. Evidence of parasympathetic hyperactivity in functionally overreached athletes. *Med Sci Sports Exerc*. 2013;45(11):2061-71.
 111. Leicht AS, Allen GD. Moderate-term reproducibility of heart rate variability during rest and light to moderate exercise in children. *Braz J Med Biol Res*. 2008;41(7):627-33.
 112. Malik M, Camm AJ, Amaral LA, Goldberger AL, Ivanov P, Stanley HE, et al. Components of heart rate variability—what they really mean and what we really measure. *Am J Cardiol*. 1993;72(11):821-2.
 113. Malone S, Owen A, Newton M, Mendes B, Collins KD, Gabbett TJ. The acute:chronic workload ratio in relation to injury risk in professional soccer. *J Sci Med Sport*. 2017;20(6):561-5.

114. Mann T, Lamberts RP, Lambert MI. Methods of prescribing relative exercise intensity: Physiological and practical considerations. *Sports Med.* 2013;43(7):613-25.
115. Manzi V, Iellamo F, Impellizzeri F, D'Ottavio S, Castagna C. Relation between individualized training impulses and performance in distance runners. *Med Sci Sports Exerc.* 2009;41(11):2090-6.
116. Manzi V, Bovenzi A, Franco Impellizzeri M, Carminati I, Castagna C. Individual training-load and aerobic-fitness variables in premier-ship soccer players during the precompetitive season. *J Strength Cond Res.* 2013;27(3):631-6.
117. Manzi V, Castagna C, Padua E, Lombardo M, D'Ottavio S, Massaro M, et al. Dose-response relationship of autonomic nervous system responses to individualized training impulse in marathon runners. *Am J Physiol Heart Circ Physiol.* 2009;296(6):H1733-40.
118. Mateo M, Blasco-Lafarga C, Martinez-Navarro I, Guzman JF, Zabala M. Heart rate variability and pre-competitive anxiety in BMX discipline. *Eur J Appl Physiol.* 2012;112(1):113-23.
119. McLean BD, Coutts AJ, Kelly V, McGuigan MR, Cormack SJ. Neuromuscular, endocrine, and perceptual fatigue responses during different length between-match microcycles in professional rugby league players. *Int J Sports Physiol Perform.* 2010;5(3):367-83.
120. McNarry MA, Lewis MJ. Heart rate variability reproducibility during exercise. *Physiol Meas.* 2012;33(7):1123-33.
121. Meeusen R, Duclos M, Gleeson M, Rietjens G, Steinacker J, Urhausen A. Prevention, diagnosis and treatment of the overtraining syndrome. *Eur J Sport Sci.* 2006;6:1-14.
122. Meeusen R, Duclos M, Foster C, Fry A, Gleeson M, Nieman D, et al. Prevention, diagnosis and treatment of the overtraining syndrome: Joint consensus statement of the European College of Sport Science (ECSS) and the American College of Sports Medicine (ACSM). *Eur J Sport Sci.* 2013;13(1):1-24.
123. Meister S, Aus der Funten K, Meyer T. Repeated monitoring of blood parameters for evaluating strain and overload in elite football players: Is it justified? *J Sports Sci.* 2014;32(13):1328-31.
124. Morales J, Garcia V, Garcia-Masso X, Salva P, Escobar R, Busca B. The use of heart rate variability in assessing precompetitive stress in high-standard judo athletes. *Int J Sports Med.* 2013;34(2):144-51.
125. Morin JB, Jeannin T, Chevallier B, Belli A. Spring-mass model characteristics during sprint running: Correlation with performance and fatigue-induced changes. *Int J Sports Med.* 2006;27(2):158-65.
126. Murray NB, Gabbett TJ, Townshend AD, Blanch P. Calculating acute:chronic workload ratios using exponentially weighted moving averages provides a more sensitive indicator of injury likelihood than rolling averages. *Br J Sports Med.* 2017;51(9):749-54.
127. Myllymaki T, Kyrolainen H, Savolainen K, Hokka L, Jakonen R, Juuti T, et al. Effects of vigorous late-night exercise on sleep quality and cardiac autonomic activity. *J Sleep Res.* 2011;20(1 Pt 2):146-53.
128. Nakamura FY, Flatt AA, Pereira LA, Ramirez-Campillo R, Loturco I, Esco MR. Ultra-short-term heart rate variability is sensitive to training effects in team sports players. *J Sports Sci Med.* 2015;14(3):602-5.
129. Norris SR, Smith DJ. Planning, periodization, and sequencing of training and competition: The rationale for a competently planned, optimally executed training and competition program, supported by a multidisciplinary team. In Kellmann M, ed. *Enhancing Recovery: Preventing Underperformance in Athletes.* Champaign, IL: Human Kinetics; 2002: 121-41.
130. Nussinovitch U, Elishkevitz KP, Katz K, Nussinovitch M, Segev S, Volovitz B, et al. Reliability of ultra-short ECG indices for heart rate variability. *Ann Noninvasive Electrocardiol.* 2011;16(2):117-22.
131. Nussinovitch U, Elishkevitz KP, Kaminer K, Nussinovitch M, Segev S, Volovitz B, et al. The efficiency of 10-second resting heart rate for the evaluation of short-term heart rate variability indices. *Pacing Clin Electrophysiol.* 2011;34(11):1498-502.
132. Otzenberger H, Gronfier C, Simon C, Charloux A, Ehrhart J, Piquard F, et al. Dynamic heart rate variability: A tool for exploring sympathovagal balance continuously during sleep in men. *Am J Physiol.* 1998;275:H946-H50.
133. Pagani M, Lucini D. Can autonomic monitoring predict results in distance runners? *Am J Physiol Heart Circ Physiol.* 2009;296(6):H1721-2.

134. Parouty J, Al Haddad H, Quod MJ, Leprêtre PM, Ahmaidi S, Buchheit M. Effect of cold water immersion on 100-m sprint performance in well-trained swimmers. *Eur J Appl Physiol*. 2010;109(3):483-90.
135. Parrado E, Garcia MA, Ramos J, Cervantes JC, Rodas G, Capdevila L. Comparison of Omega Wave System and Polar S810i to detect R-R intervals at rest. *Int J Sports Med*. 2010;31(5):336-41.
136. Penttilä J, Helminen A, Jartti T, Kuusela T, Huikuri HV, Tulppo MP, et al. Time domain, geometrical and frequency domain analysis of cardiac vagal outflow: Effects of various respiratory patterns. *Clin Physiol*. 2001;21(3):365-76.
137. Pichot V, Busso T, Roche F, Garet M, Costes F, Duverney D, et al. Autonomic adaptations to intensive and overload training periods: A laboratory study. *Med Sci Sports Exerc*. 2002;34(10):1660-6.
138. Pichot V, Roche F, Gaspoz JM, Enjolras F, Antoniadis A, Minini P, et al. Relation between heart rate variability and training load in middle-distance runners. *Med Sci Sports Exerc*. 2000;32(10):1729-36.
139. Plews DJ, Laursen PB, Kilding AE, Buchheit M. Heart rate variability in elite triathletes, is variation in variability the key to effective training? A case comparison. *Eur J Appl Physiol*. 2012;112(11):3729-41.
140. Plews DJ, Laursen PB, Kilding AE, Buchheit M. Heart rate variability and training intensity distribution in elite rowers. *Int J Sports Physiol Perform*. 2014;9(6):1026-32.
141. Plews DJ, Laursen PB, Kilding AE, Buchheit M. Evaluating training adaptation with heart rate measures: A methodological comparison. *Int J Sports Physiol Perform*. 2013;8(6):688-91.
142. Plews DJ, Laursen PB, Kilding AE, Buchheit M. Training adaptation and heart rate variability in elite endurance athletes - opening the door to effective monitoring. *Sports Med*. 2013;43(9):773-81.
143. Plews DJ, Laursen PB, Kilding AE, Buchheit M. Heart rate variability and training intensity distribution in elite rowers. *Int J Sports Physiol Perform*. 2014; Apr 3 [Epub ahead of print].
144. Plews DJ, Laursen PB, Le Meur Y, Hausswirth C, Kilding AE, Buchheit M. Monitoring training with heart rate variability: How much compliance is needed for valid assessment? *Int J Sports Physiol Perform*. 2014; In press.
145. Plews DJ, Scott B, Altini M, Wood M, Kilding AE, Laursen PB. Comparison of heart-rate-variability recording with smartphone photoplethysmography, Polar H7 Chest Strap, and electrocardiography. *Int J Sports Physiol Perform*. 2017;12(10):1324-8.
146. Portier H, Louisy F, Laude D, Berthelot, Guezennec CY. Intense endurance training on heart rate and blood pressure variability in runners. *Med Sci Sports Exerc*. 2001;33(7):1120-5.
147. Rowsell GJ, Coutts AJ, Reaburn P, Hill-Haas S. Effects of cold-water immersion on physical performance between successive matches in high-performance junior male soccer players. *J Sports Sci*. 2009;27(6):565-73.
148. Saboul D, Pialoux V, Hautier C. The impact of breathing on HRV measurements: Implications for the longitudinal follow-up of athletes. *Eur J Sport Science*. 2013; Feb 8 [Epub ahead of print].
149. Sandercock GR, Bromley PD, Brodie DA. The reliability of short-term measurements of heart rate variability. *Int J Cardiol*. 2005;103(3):238-47.
150. Saw A, Main L, Robertson S, Gastin P. Athlete self-report measure use and associated psychological alterations. *Sports*. 2017;5(3):54.
151. Saw AE, Main LC, Gastin PB. Role of a self-report measure in athlete preparation. *Journal of Strength and Conditioning Research*. 2015;29(3):685-91.
152. Saw AE, Main LC, Gastin PB. Monitoring athletes through self-report: Factors influencing implementation. *J Sports Sci Med*. 2015;14:137-46.
153. Saw AE, Main LC, Gastin PB. Monitoring the athlete training response: Subjective self-reported measures trump commonly used objective measures: A systematic review. *Br J Sports Med*. 2016;50(5):281-91.
154. Saw AE, Kellmann M, Main LC, Gastin PB. Athlete self-report measures in research and practice: Considerations for the discerning reader and fastidious practitioner. *Int J Sports Physiol Perform*. 2017;12(S2):127-35.
155. Scharhag-Rosenberger F, Meyer T, Walitzek S, Kindermann W. Time course of changes in

- endurance capacity: A 1-yr training study. *Med Sci Sports Exerc.* 2009;41(5):1130-7.
156. Schmitt L, Regnard J, Desmarests M, Mauny F, Mourot L, Fouillot JP, et al. Fatigue shifts and scatters heart rate variability in elite endurance athletes. *PLoS One.* 2013;8(8):e71588.
 157. Silva JR, Rumpf MC, Hertzog M, Castagna C, Farooq A, Girard O, et al. Acute and residual soccer match-related fatigue: A systematic review and meta-analysis. *Sports Med.* 2018;48(3):539-83.
 158. Spinelli L, Petretta M, Marciano F, Testa G, Rao MA, Volpe M, et al. Cardiac autonomic responses to volume overload in normal subjects and in patients with dilated cardiomyopathy. *Am J Physiol.* 1999;277(4 Pt 2):H1361-8.
 159. Stanley J, Peake JM, Buchheit M. Cardiac parasympathetic reactivation following exercise: Implications for training prescription. *Sports Med.* 2013;43(12):1259-77.
 160. Stanley J, Peake JM, Buchheit M. Consecutive days of cold water immersion: Effects on cycling performance and heart rate variability. *Eur J Appl Physiol.* 2013;113(2):371-84.
 161. Task Force. Heart rate variability: Standards of measurement, physiological interpretation and clinical use. Task Force of the European Society of Cardiology and the North American Society of Pacing and Electrophysiology. *Circulation.* 1996;93(5):1043-65.
 162. Taylor K, Chapman D, Cronin J, Newton M, Gill N. Fatigue monitoring in high performance sport: A survey of current trends. *Journal of Australian Strength and Conditioning.* 2012;20(1):12-23.
 163. ten Haaf T, van Staveren S, Oudenhoven E, Piacentini MF, Meeusen R, Roelands B, et al. Prediction of functional overreaching from subjective fatigue and readiness to train after only 3 days of cycling. *Int J Sports Physiol Perform.* 2017;12(S2):87-94.
 164. Thorpe RT, Strudwick AJ, Buchheit M, Atkinson G, Drust B, Gregson W. Tracking morning fatigue status across in-season training weeks in elite soccer players. *Int J Sports Physiol Perform.* 2016;11(7):947-52.
 165. Tulppo MP, Makikallio TH, Seppanen T, Airaksinen JK, Huikuri HV. Heart rate dynamics during accentuated sympathovagal interaction. *Am J Physiol.* 1998;274(3 Pt 2):H810-6.
 166. Twist C, Highton J. Monitoring fatigue and recovery in rugby league players. *Int J Sports Physiol Perform.* 2013;8(5):467-74.
 167. Wallen MB, Hasson D, Theorell T, Canlon B, Osika W. Possibilities and limitations of the Polar RS800 in measuring heart rate variability at rest. *Eur J Appl Physiol.* 2012;112(3):1153-65.
 168. Weippert M, Kumar M, Kreuzfeld S, Arndt D, Rieger A, Stoll R. Comparison of three mobile devices for measuring R-R intervals and heart rate variability: Polar S810i, Suunto t6 and an ambulatory ECG system. *Eur J Appl Physiol.* 2010;109(4):779-86.
 169. Wikipedia. http://en.wikipedia.org/wiki/Nick_Broad. Accessed June 2, 2018.
 170. Winsley RJ, Armstrong N, Bywater K, Fawcner SG. Reliability of heart rate variability measures at rest and during light exercise in children. *Br J Sports Med.* 2003;37(6):550-2.
 171. Yamamoto K, Miyachi M, Saitoh T, Yoshioka A, Onodera S. Effects of endurance training on resting and post-exercise cardiac autonomic control. *Med Sci Sports Exerc.* 2001;33(9):1496-502.
 172. Cohen S, Kamarck T, Mermelstein R. A global measure of perceived stress. *Journal of Health and Social Behavior.* 1983;24:385-96.
 173. Dean J, Whelan J, Meyers A. An incredibly quick way to assess mood states: The incredibly short POMS. Presented at Annual Conference of the Association for the Advancement of Applied Sport Psychology, 1990.
 174. Grove JR, Main LC, Partridge K, Bishop DJ, Russell S, Shepherdson A, Ferguson L. Training distress and performance readiness: Laboratory and field validation of a brief self-report measure. *Scand J Med Sci Sports.* 2014;24:e483-90.
 175. Grove JR, Prapavessis H. Preliminary evidence for the reliability and validity of an abbreviated Profile of Mood States. *International Journal of Sport Psychology.* 1992;23:93-109.
 176. Hitzschke B, Holst T, Ferrauti A, Meyer T, Pfeiffer M, Kellmann M. Entwicklung des Akutmaßes zur Erfassung von Erholung und Beanspruchung im Sport [Development of the Acute Recovery and Stress Scale]. *Diagnostica.* 2016.
 177. Hitzschke B, Kölling S, Ferrauti A, Meyer T, Pfeiffer M, Kellmann M. Entwicklung der Kurzskaala zur Erfassung von Erholung und Beanspruchung im Sport (KEB) [Development

- of the Short Recovery and Stress Scale for Sports (SSRS)]. *Zeitschrift für Sportpsychologie*. 2015;22:146-61.
178. James DV, Barnes AJ, Lopes P, Wood DM. Heart rate variability: Response following a single bout of interval training. *Int J Sports Med*. 2002;23:247-51.
 179. Kellmann M, Kallus KW. Recovery-stress questionnaire for athletes. In Kallus, KW, Kellman, M, eds. *The Recovery-Stress Questionnaires: User manual*. Frankfurt am Main: Pearson Assessment & Information GmbH; 2016:86-131.
 180. Kellmann M, Kölling S, Hitzschke B. *Das Akutmaß und die Kurzsкала zur Erfassung von Erholung und Beanspruchung im Sport: Manual [The Acute Measure and the Short Scale of Recovery and Stress for Sports: Manual]*. Köln: Sportverlag Strauß; 2016.
 181. Kellmann M, Patrick T, Botterill C, Wilson C. The Recovery-Cue and its use in applied settings: Practical suggestions regarding assessment and monitoring of recovery. In Kellmann, M, ed. *Enhancing Recovery: Preventing Underperformance in Athletes*. Champaign, IL: Human Kinetics; 2002:219-29.
 182. Kenttä G, Hassmén P, Raglin JS. Mood state monitoring of training and recovery in elite kayakers. *European Journal of Sport Science*. 2006;6:245-53.
 183. Laurent CM, Green JM, Bishop PA, Sjøkvist J, Schumacker RE, Richardson MT, Curtner-Smith M. A practical approach to monitoring recovery: Development of a perceived recovery status scale. *Journal of Strength and Conditioning Research*. 2011;25:620-28.
 184. Lundqvist C, Kenttä G. Positive emotions are not simply the absence of the negative ones: Development and validation of the Emotional Recovery Questionnaire (EmRecQ). *Sport Psychologist*. 2010;24:468-88.
 185. Main L, Grove JR. A multi-component assessment model for monitoring training distress among athletes. *European Journal of Sport Science*. 2009;9:195-202.
 186. McNair P, Lorr M, Droppleman L. *POMS Manual*. San Diego, CA: Education and Industrial Testing Service; 1981.
 187. Micklewright D, Gibson ASC, Gladwell V, Al Salman A. Development and validity of the Rating-of-Fatigue Scale. *Sports Medicine*. 2017;47:2375-93.
 188. Mourrot L, Bouhaddi M, Tordi N, Rouillon JD, Regnard J. Short- and long-term effects of a single bout of exercise on heart rate variability: Comparison between constant and interval training exercises. *Eur J Appl Physiol*. 2004;92:508-17.
 189. Niewiadomski W, Gasiorowska A, Krauss B, Mroz A, Cybulski G. Suppression of heart rate variability after supramaximal exertion. *Clin Physiol Funct Imaging*. 2007;27:309-19.
 190. Raglin JS, Morgan WP. Development of a scale for use in monitoring training-induced distress in athletes. *International Journal of Sports Medicine*. 1994;15:84-8.
 191. Rushall BS. A tool for measuring stress tolerance in elite athletes. *Journal of Applied Sport Psychology*. 1990;2:51-66.
 192. Shacham S. A shortened version of the Profile of Mood States. *Journal of Personality Assessment*. 1983;47:305-306.
 193. Terry P, Lane A, Fogarty G. Construct validity of the Profile of Mood States-Adolescents for use with adults. *Psychology of Sport and Exercise*. 2003;4:125-39.
 194. Terry PC, Lane AM, Lane HJ, Keohane L. Development and validation of a mood measure for adolescents. *Journal of Sports Sciences*. 1999;17:861-72.

Chapter 10

1. Benjamin DJ, Berger OJ, Johannesson M, Nosek AB, Wagenmakers EJ, Berk R, et al. Redefine statistical significance. *Nature Human Behaviour* [Internet]. 2017 (September 1, 2017).
2. Buchheit M. The numbers will love you back in return—I promise. *Int J Sports Physiol Perform*. 2016;11(4):551-4.
3. Buchheit M. Houston, we still have a problem. *Int J Sports Physiol Perform*. 2017;12(8):1111-14.
4. Buchheit M. Outside the Box. *Int J Sports Physiol Perform*. 2017;12(8):1001-2.
5. Buchheit M, Laursen PB. High-intensity interval training, solutions to the programming puzzle. Part II: Anaerobic energy, neuromuscular load and practical applications. *Sports Med*. 2013;43(10):927-54.
6. Buchheit M, Laursen PB. High-intensity interval training, solutions to the programming puzzle: Part I: Cardiopulmonary emphasis. *Sports Med*. 2013;43(5):313-38.

7. Close GL. Close Nutrition [Internet] 2017. (September 22) Available from: <http://www.closenutrition.com/?p=619>.
 8. Cohen J. Things I have learned (so far). *American Psychologist*. 1990;45:1304-12.
 9. Cressey E. Eric Cressey [Internet] 2017. (September, 22) Available from: <https://ericcressey.com/the-success-is-in-the-struggle>.
 10. Ericsson KA, Pool E. *Peak: Secrets From the New Science of Expertise*. Boston: Houghton Mifflin Harcourt; 2016.
 11. MacInnis MJ, Gibala MJ. Physiological adaptations to interval training and the role of exercise intensity. *J Physiol*. 2017;595(9):2915-30.
 12. Maxwell JC. *Thinking for a Change: 11 Ways Highly Successful People Approach Life and Work*. New York: Warner Books Inc.; 2005.
 13. McCaw A. *7 Keys to Being a Great Coach: Become Your Best and They Will Too*. Whyte K, ed. PrintShopCentral.com; 2016.
 14. Nuzzo R. Scientific method: Statistical errors. *Nature*. 2014;506(7487):150-2.
 15. Thompson WR. Worldwide Survey of Fitness Trends for 2018: The CREP Edition. *ACSM's Health & Fitness Journal*. 2017;21(6):10-19.
- ### Chapter 11
1. Alm P, Yu J-G. Physiological characters in mixed martial arts. *Am J Sports Sci*. 2013;1(2):12-7.
 2. Amtmann J, Amtmann K, Spath W. Lactate and rate of perceived exertion responses of athletes training for and competing in a mixed martial arts event. *J Strength Cond Res*. 2008;22:645-7.
 3. Andreato LV, Moraes SFd, Gomes TLdM, Esteves JDC, Andreato TV, Franchini E. Estimated aerobic power, muscle strength and flexibility in elite Brazilian jiu-jitsu athletes. *Science & Sports*. 2011;26(6):329-37.
 4. Atha J, Yeadon M, Sandover J, Parsons K. The damaging punch. *Br Med J*. 1985;291:1756-7.
 5. Beneke R, Beyer T, Jachner C, Erasmus J, Hutler M. Energetics of karate kumite. *Eur J Appl Physiol*. 2004;92:518-23.
 6. Brandao F, Fernandes H, Vilaca Alves J, Fonseca S, Machado Reis V. Hematological and biochemical markers after a Brazilian jiu-jitsu tournament in world-class athletes. *Rev Bras Cineantropom Desempenho Hum*. 2014;16(2):144-51.
 7. Bridge C, Ferreira da Silva Santos J, Chaabene H, Pieter W, Franchini E. Physical and physiological profiles of taekwondo athletes. *Sports Med*. 2014;44(6):713-33.
 8. Buchheit M. Monitoring training status with HR measures: Do all roads lead to Rome? *Front Physiol*. 2014;27(5):73.
 9. Buchheit M, Laursen P. High-intensity interval training, solutions to the programming puzzle. Part I: cardiopulmonary emphasis. *Sports Med*. 2013;43(5):313-38.
 10. Buchheit M, Laursen P. High-intensity interval training, solutions to the programming puzzle. Part II: anaerobic energy, neuromuscular load and practical applications. *Sports Med*. 2013;43(10):927-54.
 11. Buse G. No holds barred sport fighting: A 10 year review of mixed martial arts competition. *Br J Sports Med*. 2006;40:169-72.
 12. Callister R, Callister RJ, Staron R, Fleck S, Tesch P, Dudley G. Physiological characteristics of elite judo athletes. *Sports Med*. 1991;12(2):196-203.
 13. Chaabene H, Hachana Y, Franchini E, Mkaouer B, Chamari K. Physical and physiological profile of elite karate athletes. *Sports Med*. 2012;42(10):829-43.
 14. Chaabene H, Tabben M, Mkaouer B, Franchini E, Negra Y, Hammami M, et al. Amateur boxing: Physical and physiological attributes. *Sports Med*. 2015;45:337-52.
 15. Christen J, Foster C, Porcari J, Mikat R. Temporal robustness of the session ration of perceived exertion. *Int J Sport Phys Perf*. 2016;11(8):1088-93.
 16. Claudino J, Cronin J, Mezencio B, McMaster D, McGuigan M, Tricoli V, et al. The countermovement jump to monitor neuromuscular status: A meta-analysis. *J Sci Med Sport*. 2017;20(4):397-402.
 17. Davis P, Leithauser R, Beneke P. The energetics of semicontact 3 × 2-min amateur boxing. *Int J Sports Physiol Perform*. 2014;9(2):233-9.
 18. Del Vecchio F, Hirata S, Franchini E. A review of the time-motion analysis and combat development in mixed martial arts matches at regional level tournaments. *Perceptual & Motor Skills*. 2011;112:639-48.
 19. Estevan I, Alvarez O, Falco C, Molina-Garcia J, Castillo I. Impact force and time analysis influenced by execution distance in a round-house kick to the head in taekwondo. *J Strength Cond Res*. 2011;25(10):2851-6.

20. Franchini E, Takito M, Kiss M, Sterkowicz S. Physical fitness and anthropometrical differences between elite and non-elite judo players. *Biol Sport*. 2005;22(4):315-28.
21. Gastin P. Energy system interaction and relative contribution during maximal exercise. *Sports Med*. 2001;31(10):725-41.
22. Gathercole R, Sporer B, Stellingwerff T, Sleivert G. Comparison of the capacity of different jump and sprint field tests to detect neuromuscular fatigue. *J Strength Cond Res*. 2015;29(9):2522-31.
23. Haff G, Nimphius S. Training principles for power. *Strength Cond J*. 2012;34(6):2-12.
24. Hayes P, Quinn M. A mathematical model for quantifying training. *Eur J Appl Physiol*. 2009;106(6):839-47.
25. Heller J, Peric T, Dlouha R, Kohlikova E, Melichna J, Novakova H. Physiological profiles of male and female taekwondo (ITF) black belts. *J Sport Sci*. 1998;16:243-9.
26. Horswill C. Physiology of wrestling. In Garrett W, Kirkendall D, eds. *Exercise and Sport Science*. Philadelphia: Lippincott, Williams & Wilkins; 2000:955-64.
27. Horswill C, Scott J, Galea P. Comparison of maximum aerobic power, maximum anaerobic power, and skinfold thickness of elite and nonelite junior wrestlers. *Int J Sports Med*. 1989;10:165-8.
28. Hulin B, Gabbett T, Caputi P, Lawson D, Sampson J. Low chronic workload and the acute:chronic workload ratio are more predictive of injury than between-match recovery time: A two-season prospective cohort study in elite rugby league players. *Br J Sports Med*. 2016;50(16):1008-12.
29. Issurin V. Block periodization versus traditional training theory: A review. *J Sports Med Phys Fitness*. 2008;48(1):65-75.
30. James L, Haff G, Kelly V, Beckman E. Towards a determination of the physiological characteristics distinguishing successful mixed martial arts athletes: A systematic review of combat sport literature. *Sports Med*. 2016;46:1525-51.
31. James L, Beckman E, Kelly V, Haff G. The neuromuscular qualities of higher and lower-level mixed martial arts competitors. *Int J Sport Phys Perf*. 2016;12(5):612-20.
32. James L, Robertson S, Haff G, Beckman E, Kelly V. Identifying the performance characteristics of a winning outcome in elite mixed martial arts competition. *J Sci Med Sport*. 2017;20(3):296-301.
33. Jamieson J. *Ultimate MMA Conditioning*. Kirkland, WA: Performance Sports, Inc.; 2009.
34. Khanna G, Manna I. Study of the physiological profile of Indian boxers. *J Sports Sci Med*. 2006;5:90-8.
35. Kraemer W, Fry A, Rubin M, Triplett-McBride T, Gordon S, Koziris L, et al. Physiological and performance responses to tournament wrestling. *Med Sci Sports Exerc*. 2001;33(8):1367-78.
36. Laursen P, Jenkins D. The scientific basis for high-intensity interval training. *Sports Med*. 2002;32(1):53-73.
37. Lellamo F, Pigozzi F, Spataro A, Lucini D, Pagani M. T-wave and heart rate variability changes to assess training in world-class athletes. *Med Sci Sports Exerc*. 2004;36(8):1342-6.
38. MacDougall D, Sale D. *The Physiology of Training for High Performance*. Oxford: Oxford University Press; 2014.
39. Mack J, Stojisih S, Sherman S, Dau N, Bir C, eds. Amateur boxer biomechanics and punch force. In Jensen R, Ebben W, Petushek E, Richter C, Roemer K, eds. *Proceedings of the 28th Conference of the International Society of Biomechanics in Sports*. Marquette, MI: n.p., 2010:453-56.
40. Matsushigue K, Hartmann K, Franchini E. Taekwondo: Physiological responses and match analysis. *J Strength Cond Res*. 2009;23:1112-7.
41. Miarka B, Coswig V, Vecchio F, Brito C, Amtmann J. Comparisons of time-motion analysis of mixed martial arts rounds by weight division. *Int J Perf Anal Sport*. 2015;15:1189-201.
42. Murray N, Gabbett T, Townsend A, Blanch P. Calculating acute:chronic workload ratios using exponentially weighted moving averages provides a more sensitive indicator of injury likelihood than rolling averages. *Br J Sports Med*. 2016;51(9):749-54.
43. Nilsson J, Csargo S, Gullstrand L, Tveit P, Refsnes P. Work-time profile, blood lactate concentration, and rating of perceived exertion in the 1998 Greco-Roman wrestling World Championship. *J Sport Sci*. 2002;20:939-45.
44. Oliveira M, Moreira D, Godoy J, Cambraia A. Avaliacao da forca de preensao palmar em atletas

- de jiu-jitsu de nivel competitivo. *Rev Bras Ciencia e Movi.* 2006;14(3):63-70.
45. Pedzich W, Mastalera A, Urbanik C. The comparison of the dynamics of selected leg strokes in taekwondo WTF. *Acta Bioeng Biomech.* 2006;8(1):1-9.
 46. Saw A, Main L, Gastin P. Monitoring the athlete training response: Subjective self-reported measures trump commonly used objective measures: A systematic review. *Sports Med.* 2016;50(5):281-91.
 47. Schick M, Brown L, Coburn J, Beam W, Schick E, Dabbs N. Physiological profile of mixed martial artists. *Med Sport.* 2010;14(4):182-7.
 48. Smith M. Physiological profile of senior and junior England international amateur boxers. *J Sports Sci Med.* 2006;5:74-89.
 49. Song T, Garvie G. Anthropometric, flexibility, strength and physiological measures of Canadian wrestlers and comparison of Canadian and Japanese Olympic wrestlers. *Can J Appl Sport Sci.* 1980;5:1-8.
 50. Tabata I, Irisawa K, Kouzaki M, Nishimura K, Ogita F, Miyachi M. Metabolic profile of high intensity intermittent exercise. *Med Sci Sports Exerc.* 1997;29(3):390-5.
 51. Vecchio FD, Bianchi S, Hirata S, Chacon-Mikahil M. Analise morfo-funcional de praticantes de brazilian jiu-jitsu e estudo da temporalidade e da quantificação das ações motoras na modalidade. *Movimento e Percepção.* 2007;7:263-81.
 52. Walilko T, Viano D, Bir C. Biomechanics of the head for Olympic boxer punches to the face. *Br J Sports Med.* 2005;39(10):710-9.
 53. Yoon J. Physiological profiles of elite senior wrestlers. *Sports Med.* 2002;32(4):225-33.
 54. Yoon J, Bang D, Jun H. The development of sparring types for elite Korean national wrestlers. *Korean J Sport Sci.* 1994;5(2):15-24.
 3. Andersson E, Holmberg HC, Ortenblad N, Bjorklund G. Metabolic responses and pacing strategies during successive sprint skiing time trials. *Med Sci Sports Exerc.* 2016.
 4. Andersson E, Pellegrini B, Sandbakk O, Stuggl T, Holmberg HC. The effects of skiing velocity on mechanical aspects of diagonal cross-country skiing. *Sports Biomech.* 2014;13(3):267-84.
 5. Andersson E, Supej M, Sandbakk O, Sperlich B, Stoggl T, Holmberg HC. Analysis of sprint cross-country skiing using a differential global navigation satellite system. *Eur J Appl Physiol.* 2010;110(3):585-95.
 6. Bolger CM, Kocbach J, Hegge AM, Sandbakk O. Speed and heart-rate profiles in skating and classical cross-country skiing competitions. *Int J Sports Physiol Perform.* 2015;10(7):873-80.
 7. Helgerud J, Hoydal K, Wang E, Karlsen T, Berg P, Bjerkaas M, et al. Aerobic high-intensity intervals improve VO₂max more than moderate training. *Med Sci Sports Exerc.* 2007;39(4):665-71.
 8. Holmberg HC. The elite cross-country skier provides unique insights into human exercise physiology. *Scand J Med Sci Sports.* 2015;25 Suppl 4:100-9.
 9. Holmberg HC, Rosdahl H, Svedenhag J. Lung function, arterial saturation and oxygen uptake in elite cross country skiers: influence of exercise mode. *Scand J Med Sci Sports.* 2007;17(4):437-44.
 10. Lindinger SJ, Stoggl T, Muller E, Holmberg HC. Control of speed during the double poling technique performed by elite cross-country skiers. *Med Sci Sports Exerc.* 2009;41(1):210-20.
 11. Losnegard T, Hallen J. Physiological differences between sprint- and distance-specialized cross-country skiers. *Int J Sports Physiol Perform.* 2014;9(1):25-31.
 12. Losnegard T, Myklebust H, Hallen J. Anaerobic capacity as a determinant of performance in sprint skiing. *Med Sci Sports Exerc.* 2012;44(4):673-81.
 13. Losnegard T, Kjeldsen K, Skattebo O. An analysis of the pacing strategies adopted by elite cross-country skiers. *J Strength Cond Res.* 2016.
 14. Losnegard T, Myklebust H, Spencer M, Hallen J. Seasonal variations in VO₂max, O₂-cost, O₂-deficit, and performance in elite cross-country skiers. *J Strength Cond Res.* 2013;27(7):1780-90.

Chapter 12

15. Norman RW, Ounpuu S, Fraser M, Mitchell R. Mechanical power output and estimated metabolic rates of Nordic skiers during Olympic competition. *Int J Sport Biomech.* 1989;5:169-84.
16. Ortenblad N, Westerblad H, Nielsen J. Muscle glycogen stores and fatigue. *J Physiol.* 2013;591(18):4405-13.
17. Sandbakk O, Holmberg HC. A reappraisal of success factors for Olympic cross-country skiing. *Int J Sports Physiol Perform.* 2014;9(1):117-21.
18. Sandbakk O, Holmberg HC. Physiological capacity and training routines of elite cross-country skiers: Approaching the upper limits of human endurance. *Int J Sports Physiol Perform.* 2017:1-26.
19. Sandbakk O, Ettema G, Holmberg HC. The influence of incline and speed on work rate, gross efficiency and kinematics of roller ski skating. *Eur J Appl Physiol.* 2012;112(8):2829-38.
20. Sandbakk O, Hegge AM, Ettema G. The role of incline, performance level, and gender on the gross mechanical efficiency of roller ski skating. *Front Physiol.* 2013;4:293.
21. Sandbakk O, Holmberg HC, Leirdal S, Ettema G. Metabolic rate and gross efficiency at high work rates in world class and national level sprint skiers. *Eur J Appl Physiol.* 2010;109(3):473-81.
22. Sandbakk O, Holmberg HC, Leirdal S, Ettema G. The physiology of world-class sprint skiers. *Scand J Med Sci Sports.* 2011;21(6):e9-16.
23. Sandbakk O, Sandbakk SB, Ettema G, Welde B. Effects of intensity and duration in aerobic high-intensity interval training in highly trained junior cross-country skiers. *J Strength Cond Res.* 2013;27(7):1974-80.
24. Sandbakk O, Ettema G, Leirdal S, Jakobsen V, Holmberg HC. Analysis of a sprint ski race and associated laboratory determinants of world-class performance. *Eur J Appl Physiol.* 2011;111(6):947-57.
25. Sandbakk O, Hegge AM, Losnegard T, Skattebo O, Tonnessen E, Holmberg HC. The physiological capacity of the world's highest ranked female cross-country skiers. *Med Sci Sports Exerc.* 2016;48(6):1091-100.
26. Sandbakk O, Losnegard T, Skattebo O, Hegge AM, Tonnessen E, Kocbach J. Analysis of classical time-trial performance and technique-specific physiological determinants in elite female cross-country skiers. *Front Physiol.* 2016;7:326.
27. Stoggl T, Holmberg HC. Double-poling biomechanics of elite cross-country skiers: Flat versus uphill terrain. *Med Sci Sports Exerc.* 2016.
28. Stoggl T, Lindinger S, Muller E. Evaluation of an upper-body strength test for the cross-country skiing sprint. *Med Sci Sports Exerc.* 2007;39(7):1160-9.
29. Stoggl T, Muller E, Ainegren M, Holmberg HC. General strength and kinetics: Fundamental to sprinting faster in cross country skiing? *Scand J Med Sci Sports.* 2011;21(6):791-803.
30. Sylta O, Tonnessen E, Seiler S. Do elite endurance athletes report their training accurately? *Int J Sports Physiol Perform.* 2014;9(1):85-92.
31. Sylta O, Tonnessen E, Sandbakk O, Hammarstrom D, Danielsen J, Skovereng K, et al. Effects of HIIT on physiological and hormonal adaptations in well-trained cyclists. *Med Sci Sports Exerc.* 2017;49(6):1137-46.
32. Tonnessen E, Haugen TA, Hem E, Leirstein S, Seiler S. Maximal aerobic capacity in the winter-Olympics endurance disciplines: Olympic-medal benchmarks for the time period 1990-2013. *Int J Sports Physiol Perform.* 2015;10(7):835-9.
33. Vandbakk K, Welde B, Kruken AH, Baumgart J, Ettema G, Karlsen T, et al. Effects of upper-body sprint-interval training on strength and endurance capacities in female cross-country skiers. *PLoS One.* 2017;12(2):e0172706.

Chapter 13

1. Arrese AL, Izquierdo DM, Urdiales DM. A review of the maximal oxygen uptake values necessary for different running performance levels. *New Studies in Athletic.* 2005;20(3):7-20.
2. Boileau RA, Mayhew JL, Riner WF, Lussier L. Physiological characteristics of elite middle and long distance runners. *Can J Appl Sport Sci.* 1982;7(3):167-72.
3. Brandon LJ. Physiological factors associated with middle distance running performance. *Sports Med.* 1995;19(4):268-77.
4. Coe P. *Winning Running: Successful 800m & 1500m Racing and Training*: The Crowood Press, Ramsbury, Marlborough, Wiltshire; 1996.
5. Duffield R, Dawson B, Goodman C. Energy system contribution to 1500- and 3000-metre

- track running. *J Sports Sci.* 2005;23(10):993-1002.
6. Hanon C, Lehenaff D, Gajer B, Mathieu C, Vollmer JC. Analyse comparée de différentes séances de développement de VO2max. *Il èmes journées internationales des sciences du sport: Expertise et sport de haut-niveau*, November 12-15, 2002.
 7. Hanon C, Thomas C. Effects of optimal pacing strategies for 400-, 800-, and 1500-m races on the VO2 response. *J Sports Sci.* 2011;29(9):905-12.
 8. Hanon C, Leveque JM, Thomas C, Vivier L. Pacing strategy and VO2 kinetics during a 1500-m race. *Int J Sports Med.* 2008;29(3):206-11.
 9. Houmard JA, Costill DL, Mitchell JB, Park SH, Chenier TC. The role of anaerobic ability in middle distance running performance. *Eur J Appl Physiol Occup Physiol.* 1991;62(1):40-3.
 10. Ingham SA, Whyte GP, Pedlar C, Bailey DM, Dunman N, Nevill AM. Determinants of 800-m and 1500-m running performance using allometric models. *Med Sci Sports Exerc.* 2008;40(2):345-50.
 11. Keul J. Die Bedeutung des aeroben und anaeroben Leistungsvermögens für Mittel und Langstreckenläufer *Die Lehre der Leichtathletik* 1975:7-18.
 12. Noakes TD, Lambert MI, Hauman R. Which lap is the slowest? An analysis of 32 world mile record performances. *Br J Sports Med.* 2009;43(10):760-4.
 13. Rabadan M, Diaz V, Calderon FJ, Benito PJ, Peinado AB, Maffulli N. Physiological determinants of specialty of elite middle- and long-distance runners. *J Sports Sci.* 2011;29(9):975-82.
 14. Spencer MR, Gastin PB. Energy system contribution during 200- to 1500-m running in highly trained athletes. *Med Sci Sports Exerc.* 2001;33(1):157-62.
 15. Svedenhag J, Sjodin B. Maximal and submaximal oxygen uptakes and blood lactate levels in elite male middle- and long-distance runners. *Int J Sports Med.* 1984;5(5):255-61.
 16. Thiel C, Foster C, Banzer W, De Koning J. Pacing in Olympic track races: Competitive tactics versus best performance strategy. *J Sports Sci.* 2012;30(11):1107-15.
 17. Tjelta LI. Physiological factors affecting performance in elite distance runners. *Acta Kinesiologiae Universitatis Tartuensis.* 2016;22:7-19.
 18. Weyand PG, Davis JA. Running performance has a structural basis. *J Exp Biol.* 2005;208(Pt 14):2625-31.
 19. Wilson H. *Running My Way*: N.p.: Sackville Books; 1998.
- ## Chapter 14
1. Banister EW, Calvert TW. Planning for future performance: Implications for long term training. *Canadian Journal of Applied Sport Sciences Journal Canadien des Sciences Appliquees au Sport.* 1980;5(3):170-6.
 2. Barnes K, Hopkins W, McGuigan M, Kilding A. Effects of different uphill interval-training programs on running economy and performance. *Int J Sports Physiol Perform.* 2013;8(6):639-47.
 3. Barnes KR, Kilding AE. Strategies to improve running economy. *Sports Med.* 2015;45(1):37-56.
 4. Bartlett JD, Hawley JA, Morton JP. Carbohydrate availability and exercise training adaptation: Too much of a good thing? *Eur J Sport Sci.* 2015;15(1):3-12.
 5. Billat V. Interval training for performance: A scientific and empirical practice. Special recommendations for middle- and long-distance running. Part II: Anaerobic interval training. *Sports Med.* 2001;31(2):75-90.
 6. Billat V. Interval training for performance: A scientific and empirical practice. Special recommendations for middle- and long-distance running. Part I: Aerobic interval training. *Sports Med.* 2001;31(1):13-31.
 7. Billat V, Lepretre PM, Heugas AM, Laurence MH, Salim D, Koralsztejn JP. Training and bioenergetic characteristics in elite male and female Kenyan runners. *Med Sci Sports Exerc.* 2003;35(2):297-304; discussion 5-6.
 8. Billat VL, Demarle A, Slawinski J, Paiva M, Koralsztejn JP. Physical and training characteristics of top-class marathon runners. *Med Sci Sports Exerc.* 2001;33(12):2089-97.
 9. Bonetti DL, Hopkins WG. Sea-level exercise performance following adaptation to hypoxia: A meta-analysis. *Sports Med.* 2009;39(2):107-27.
 10. Buchheit M. Monitoring training status with HR measures: Do all roads lead to Rome? *Frontiers in Physiology.* 2014;5:73.
 11. Buchheit M, Laursen PB. High-intensity interval training, solutions to the programming puzzle.

- Part I: Cardiopulmonary emphasis. *Sports Med.* 2013;43(5):313-38.
12. Buchheit M, Laursen PB. High-intensity interval training, solutions to the programming puzzle. Part II: Anaerobic energy, neuromuscular load and practical applications. *Sports Med.* 2013;43(10):927-54.
 13. Burke LM, Read RS. Dietary supplements in sport. *Sports Medicine.* 1993;15(1):43-65.
 14. Casadio JR, Kilding AE, Cotter JD, Laursen PB. From lab to real world: Heat acclimation considerations for elite athletes. *Sports Med.* 2016.
 15. Chapman DW, Sharma AP, Saunders AG. The effect of brief downhill running training on running economy and lower body strength and power. *J Strength Cond Res.* 2017;31(Suppl 1):S115-16.
 16. Close GL, Hamilton DL, Philp A, Burke LM, Morton JP. New strategies in sport nutrition to increase exercise performance. *Free Radic Biol Med.* 2016;98:144-58.
 17. Esteve-Lanao J, Foster C, Seiler S, Lucia A. Impact of training intensity distribution on performance in endurance athletes. *J Strength Cond Res.* 2007;21(3):943-9.
 18. Esteve-Lanao J, San Juan AF, Earnest CP, Foster C, Lucia A. How do endurance runners actually train? Relationship with competition performance. *Med Sci Sports Exerc.* 2005;37(3):496-504.
 19. Foster C, Rodriguez-Marroyo JA, de Koning JJ. Monitoring training loads: The past, the present, and the future. *Int J Sports Physiol Perform.* 2017;12(Suppl 2):S22-S8.
 20. Foster C, Florhaug JA, Franklin J, Gottschall L, Hrovatin LA, Parker S, et al. A new approach to monitoring exercise training. *J Strength Cond Res.* 2001;15(1):109-15.
 21. Franch J, Madsen K, Djurhuus MS, Pedersen PK. Improved running economy following intensified training correlates with reduced ventilatory demands. *Med Sci Sports Exerc.* 1998;30(8):1250-6.
 22. Gabbett TJ. The training-injury prevention paradox: Should athletes be training smarter and harder? *Br J Sports Med.* 2016;50(5):273-80.
 23. Gabbett TJ, Ullah S. Relationship between running loads and soft-tissue injury in elite team sport athletes. *J Strength Cond Res.* 2012;26(4):953-60.
 24. Garrett A, Creasy R, Rehrer N, Patterson M, Cotter J. Effectiveness of short-term heat acclimation for highly trained athletes. *Eur J Appl Physiol.* 2012;112(5):1827-37.
 25. Gore CJ, Clark SA, Saunders PU. Nonhematological mechanisms of improved sea-level performance after hypoxic exposure. *Med Sci Sports Exerc.* 2007;39(9):1600-9.
 26. Gore CJ, Sharpe K, Garvican-Lewis LA, Saunders PU, Humberstone CE, Robertson EY, et al. Altitude training and haemoglobin mass from the optimised carbon monoxide rebreathing method determined by a meta-analysis. *Br J Sports Med.* 2013;47 Suppl 1:i31-9.
 27. Hetlelid KJ, Plews DJ, Herold E, Laursen PB, Seiler S. Rethinking the role of fat oxidation: Substrate utilisation during high-intensity interval training in well-trained and recreationally trained runners. *BMJ Open Sport & Exercise Medicine.* 2015;1(1).
 28. Issurin V. New horizons for the methodology and physiology of training periodization. *Sports Med.* 2010;1(40):189-206.
 29. Jeukendrup AE. Periodized nutrition for athletes. *Sports Med.* 2017;1-13.
 30. Kruse TN, Carter RE, Rosedahl JK, Joyner MJ. Speed trends in male distance running. *PLoS ONE.* 2014;9(11):e112978.
 31. Larsen HB, Sheel AW. The Kenyan runners. *Scand J Med Sci Sports.* 2015;25 Suppl 4:110-8.
 32. Laursen P, Jenkins D. The scientific basis for high-intensity interval training: Optimising training programmes and maximising performance in highly trained endurance athletes. *Sports Med.* 2002;32(1):53-73.
 33. Laursen PB. Training for intense exercise performance: High-intensity or high-volume training? *Scand J Med Sci Sports.* 2010;20:1-10.
 34. Le Meur Y, Hausswirth C, Natta F, Couturier A, Bignet F, Vidal PP. A multidisciplinary approach to overreaching detection in endurance trained athletes. *J Appl Physiol (1985).* 2013;114(3):411-20.
 35. Lobigs LM, Sharpe K, Garvican-Lewis LA, Gore CJ, Peeling P, Dawson B, et al. The athlete's hematological response to hypoxia: A meta-analysis on the influence of altitude exposure on key biomarkers of erythropoiesis. *Am J Hematol.* 2018;93(1):74-83.

36. Lundby C, Robach P. Does 'altitude training' increase exercise performance in elite athletes? *Exp Physiol*. 2016;101(7):783-88.
37. Maffetone PB, Malcata R, Rivera I, Laursen PB. The Boston Marathon versus the World Marathon Majors. *PLoS ONE*. 2017;12(9):e0184024.
38. Marquet LA, Brisswalter J, Louis J, Tiollier E, Burke LM, Hawley JA, et al. Enhanced endurance performance by periodization of CHO intake: "Sleep low" strategy. *Med Sci Sports Exerc*. 2016;48(4):663-72.
39. McMahon NF, Leveritt MD, Pavey TG. The effect of dietary nitrate supplementation on endurance exercise performance in healthy adults: A systematic review and meta-analysis. *Sports Med*. 2017;47(4):735-56.
40. Mytton GJ, Archer DT, Turner L, Skorski S, Renfree A, Thompson KG, et al. Increased variability of lap speeds: Differentiating medalists and nonmedalists in middle-distance running and swimming events. *Int J Sports Physiol Perform*. 2015;10(3):369-73.
41. Paluska S. Caffeine and exercise. *Curr Sports Med Rep*. 2003;2(4):213-9.
42. Periard JD, Racinais S, Sawka MN. Adaptations and mechanisms of human heat acclimation: Applications for competitive athletes and sports. *Scand J Med Sci Sports*. 2015;25 Suppl 1:20-38.
43. Racinais S, Alonso JM, Coutts AJ, Flouris AD, Girard O, Gonzalez-Alonso J, et al. Consensus recommendations on training and competing in the heat. *Scand J Med Sci Sports*. 2015;25 Suppl 1:6-19.
44. Robinson N, Sottas PE, Schumacher YO. The athlete biological passport: How to personalize anti-doping testing across an athlete's career? *Med Sport Sci*. 2017;62:107-18.
45. Ronnestad BR, Hansen J. A scientific approach to improve physiological capacity of an elite cyclist. *Int J Sports Physiol Perform*. 2017;1-11.
46. Ronnestad BR, Hansen J, Ellefsen S. Block periodization of high-intensity aerobic intervals provides superior training effects in trained cyclists. *Scand J Med Sci Sports*. 2014;24(1):34-42.
47. Ronnestad BR, Hansen J, Thyli V, Bakken TA, Sandbakk O. 5-week block periodization increases aerobic power in elite cross-country skiers. *Scand J Med Sci Sports*. 2016;26(2):140-46.
48. Sandford GN, Pearson S, Allen SV, Malcata RM, Kilding AE, Ross A, et al. Tactical behaviors in men's 800m Olympic and World Championship medalists: A changing of the guard. *Int J Sports Physiol Perform*. 2018;13(2):246-49.
49. Saw AE, Main LC, Gastin PB. Monitoring the athlete training response: Subjective self-reported measures trump commonly used objective measures: A systematic review. *Br J Sports Med*. 2016;50(5):281-91.
50. Seiler KS. What is best practice for training intensity and duration distribution in endurance athletes? *Int J Sports Physiol Perform*. 2010;5(3):276-91.
51. Seiler KS. Seiler's hierarchy of endurance training needs. *European Endurance Conference, European Athletics Coaching Summit Series*. 2016. Available online at: www.researchgate.net/publication/310725768_Seiler's_Hierarchy_of_Endurance_Training_Needs (accessed September 9, 2017).
52. Sharma AP, Garvican-Lewis LA, Clark B, Stanley J, Robertson EY, Saunders PU, et al. Training at 2100 m altitude affects running speed and session RPE at different intensities in elite middle-distance runners. *Int J Sports Physiol Perform*. 2017;12(Suppl 2): S2147-S2152.
53. Sjodin B, Svedenhag J. Applied physiology of marathon running. *Sports Med*. 1985;2(2):83-99.
54. Skiba PF. Calculation of Power Output and Quantification of Training Stress in Distance Runners: The Development of the GOVSS Algorithm. 2006. Available online at: <http://runscribe.com/wp-content/uploads/power/GOVSS.pdf> (accessed September 9, 2017).
55. Stanley J, Peake J, Buchheit M. Cardiac parasympathetic reactivation following exercise: Implications for training prescription. *Sports Med*. 2013;43(12):1259-77.
56. Stanley J, D'Auria S, Buchheit M. Cardiac parasympathetic activity and race performance: an elite triathlete case study. *Int J Sports Physiol Perform*. 2015;10(4):528-34.
57. Stanley J, Halliday A, D'Auria S, Buchheit M, Leicht A. Effect of sauna-based heat acclimation on plasma volume and heart rate variability. *Eur J Appl Physiol*. 2015;115(4):785-94.
58. Stellingwerff T. Case study: Nutrition and training periodization in three elite marathon runners. *Int J Sport Nutr Exerc Metab*. 2012;22(5):392-400.

59. Tonnessen E, Sylta O, Haugen TA, Hem E, Svendsen IS, Seiler S. The road to gold: training and peaking characteristics in the year prior to a gold medal endurance performance. *PLoS One*. 2014;9(7):e101796.
 60. Tonnessen E, Svendsen IS, Ronnestad BR, Hisdal J, Haugen TA, Seiler S. The annual training periodization of 8 world champions in orienteering. *Int J Sports Physiol Perform*. 2015;10(1):29-38.
 61. Wilber RL, Pitsiladis YP. Kenyan and Ethiopian distance runners: What makes them so good? *Int J Sports Physiol Perform*. 2012;7(2):92-102.
- ### Chapter 15
1. Abbiss CR, Quod MJ, Levin G, Martin DT, Laursen PB. Accuracy of the Velotron ergometer and SRM power meter. *Int J Sports Med*. 2009;30(2):107-12.
 2. Abbiss CR, Straker L, Quod MJ, Martin DT, Laursen PB. Examining pacing profiles in elite female road cyclists using exposure variation analysis. *Br J Sports Med*. 2010;44(6):437-42.
 3. Allen H, Coggan A. *Training and Racing With a Power Meter*: Boulder, CO: VeloPress; 2010.
 4. Bell PG, Furber MJ, KA VANS, Anton-Solanas A, Swart J. The physiological profile of a multiple Tour de France winning cyclist. *Med Sci Sports Exerc*. 2017;49(1):115-23.
 5. Burke LM. Nutritional practices of male and female endurance cyclists. *Sports Med*. 2001;31(7):521-32.
 6. Close GL, Hamilton DL, Philp A, Burke LM, Morton JP. New strategies in sport nutrition to increase exercise performance. *Free Radic Biol Med*. 2016;98:144-58.
 7. Ebert TR, Martin DT, Stephens B, Withers RT. Power output during a professional men's road-cycling tour. *Int J Sports Physiol Perform*. 2006;1(4):324-35.
 8. Gardner AS, Stephens S, Martin DT, Lawton E, Lee H, Jenkins D. Accuracy of SRM and power tap power monitoring systems for bicycling. *Med Sci Sports Exerc*. 2004;36(7):1252-8.
 9. Hansen EA, Jorgensen LV, Jensen K, Fregly BJ, Sjogaard G. Crank inertial load affects freely chosen pedal rate during cycling. *J Biomech*. 2002;35(2):277-85.
 10. Hawley JA, Stepto NK. Adaptations to training in endurance cyclists: implications for performance. *Sports Med*. 2001;31(7):511-20.
 11. Impey SG, Hearris MA, Hammond KM, Bartlett JD, Louis J, Close GL, et al. Fuel for the work required: A theoretical framework for carbohydrate periodization and the glycogen threshold hypothesis. *Sports Med*. 2018; 48(5):1031-48.
 12. Laursen PB. Training for intense exercise performance: High-intensity or high-volume training? *Scan J Med Sci Sports*. 2010;20 Suppl 2:1-10.
 13. Lucia A, Hoyos J, Chicharro JL. Physiology of professional road cycling. *Sports Med*. 2001;31(5):325-37.
 14. Lucia A, Pardo J, Duran A, Hoyos J, Chicharro JL. Physiological differences between professional and elite road cyclists. *Int J Sports Med*. 1998;19(5):342-8.
 15. Matveyev L. *Fundamentals of Sports Training*. Moscow: Progress; 1981.
 16. Menaspá P, Sias M, Bates G, La Torre A. Demands of World Cup competitions in elite women's road cycling. *Int J Sports Physiol Perform*. 2017;12(10):1293-6.
 17. Menaspá P, Quod M, Martin DT, Peiffer JJ, Abbiss CR. Physical demands of sprinting in professional road cycling. *Int J Sports Med*. 2015;36(13):1058-62.
 18. Padilla S, Mujika I, Orbananos J, Santisteban J, Angulo F, Jose Goiriena J. Exercise intensity and load during mass-start stage races in professional road cycling. *Med Sci Sports Exerc*. 2001;33(5):796-802.
 19. Passfield L, Hopker JG, Jobson S, Friel D, Zabala M. Knowledge is power: Issues of measuring training and performance in cycling. *J Sports Sci*. 2017;35(14):1426-34.
 20. Quod MJ, Martin DT, Martin JC, Laursen PB. The power profile predicts road cycling MMP. *Int J Sports Med*. 2010;31(6):397-401.
 21. Stone MH, Stone M., Sands, W.A. *Principles and Practice of Resistance Training*. Champaign, IL: Human Kinetics; 2007.
 22. Tschien P. A necessary direction in training: The integration of biological adaptation in the training program. *Coaching and Sport Science Journal*. 1995;1:2-14.
- ### Chapter 16
1. Banister EW, editor. *Modeling Elite Athletic Performance*: Champaign, IL: Human Kinetics; 1991.

2. Foster C, Florhaug JA, Franklin J, Gottschall L, Hrovatin LA, Parker S, et al. A new approach to monitoring exercise training. *J Strength Cond Res.* 2001;15(1):109-15.
 3. Hagerman FC. Applied physiology of rowing. *Sports Med.* 1984;1(4):303-26.
 4. Holt AC, Plews D, Oberlin-Brown K, Merien F, Kilding AE. Autonomic and performance recovery after high intensity exercise in rowers. *Int J Sport Physiol Perform.* 2017;In Press.
 5. Ingham SA, Whyte GP, Jones K, Nevill AM. Determinants of 2,000 m rowing ergometer performance in elite rowers. *Eur J Appl Physiol.* 2002;88(3):243-6.
 6. Issurin V. Block periodization versus traditional training theory: A review. *J Sports Med Phys Fitness.* 2008;48(1):65-75.
 7. Kerr DA, Ross WD, Norton K, Hume P, Kagawa M, Ackland TR. Olympic lightweight and open-class rowers possess distinctive physical and proportionality characteristics. *J Sports Sci.* 2007;25(1):43-53.
 8. Kleshnev V, ed. *The Biomechanics of Rowing.* United Kingdom: The Crowood Press; 2016.
 9. Laursen PB, Jenkins DG. The scientific basis for high-intensity interval training: Optimising training programmes and maximising performance in highly trained endurance athletes. *Sports Med.* 2002;32(1):53-73.
 10. Lawton TW, Cronin JB, McGuigan MR. Strength testing and training of rowers: A review. *Sports Med.* 2011;41(5):413-32.
 11. Lawton TW, Cronin JB, McGuigan MR. Strength, power, and muscular endurance exercise and elite rowing ergometer performance. *J Strength Cond Res.* 2013;27(7):1928-35.
 12. Maestu J, Jurimae J, Jurimae T. Monitoring of performance and training in rowing. *Sports Med.* 2005;35(7):597-617.
 13. Marcolin G, Lentola A, Paoli A, and Petrone N. Rowing on a boat versus rowing on an ergometer: A biomechanical and electromyographical preliminary study. *Procedia Engineering* 2015;112 461-6.
 14. Neal CM, Hunter AM, Galloway SD. A 6-month analysis of training-intensity distribution and physiological adaptation in Ironman triathletes. *J Sports Sci.* 2011;29(14):1515-23.
 15. Neal CM, Hunter AM, Brennan L, O'Sullivan A, Hamilton DL, De Vito G, et al. Six weeks of a polarized training-intensity distribution leads to greater physiological and performance adaptations than a threshold model in trained cyclists. *J Appl Physiol (1985).* 2013;114(4):461-71.
 16. Plews DJ, Laursen PB. Training intensity distribution over a four-year cycle in Olympic champion rowers: Different roads lead to Rio. *Int J Sport Physiol.* 2017;27(9):1-24.
 17. Plews DJ, Laursen PB, Stanley J, Kilding AE, Buchheit M. Training adaptation and heart rate variability in elite endurance athletes: Opening the door to effective monitoring. *Sports Med.* 2013;43(9):773-81.
 18. Samuels CH. Jet lag and travel fatigue: A comprehensive management plan for sport medicine physicians and high-performance support teams. *Clin J Sport Med.* 2012;22(3):268-73.
 19. Seiler S. What is best practice for training intensity and duration distribution in endurance athletes? *Int J Sports Physiol Perform.* 2010;5(3):276-91.
 20. Stanley J, Peake JM, Buchheit M. Cardiac parasympathetic reactivation following exercise: Implication for training prescription. *Sport Med.* 2013; 43(12):1259-77.
 21. Steinacker JM. Physiological aspects of training in rowing. *Int J Sports Med.* 1993;14 Suppl 1:S3-10.
 22. Stoggl TL, Sperlich B. The training intensity distribution among well-trained and elite endurance athletes. *Front Physiol.* 2015;6:295.
- ## Chapter 17
1. Achten J, Jeukendrup AE. Heart rate monitoring. *Sports Med.* 2003;33(7):517-38.
 2. Anderson M, Hopkins W, Roberts A, Pyne D. Ability of test measures to predict competitive performance in elite swimmers. *J Sports Sci.* 2008;26(2):123-30.
 3. Anderson ME, Hopkins WG, Roberts AD, Pyne DB. Monitoring seasonal and long-term changes in test performance in elite swimmers. *Eur J Sport Sci.* 2006;6(3):145-54.
 4. Bex T, Baguet A, Achten E, Aerts P, De Clercq D, Derave W. Cyclic movement frequency is associated with muscle typology in athletes. *Scand J Med Sci Sports.* 2017;27(2):223-9.

5. Borg GA. Psychophysical bases of perceived exertion. *Med Sci Sports Exerc.* 1982;14(5):377-81.
6. Buchheit M. Outside the box. *Int J Sports Physiol Perform.* 2017;12:1001-2.
7. Carlile F. *Forbes Carlile on Swimming.* London: Pelham Books; 1963.
8. Chatard J-C, Wilson B. Effect of fastskin suits on performance, drag, and energy cost of swimming. *Med Sci Sports Exerc.* 2008;40(6):1149.
9. Costill D, Thomas R, Robergs R, Pascoe D, Lambert C, Barr S, et al. Adaptations to swimming training: Influence of training volume. *Med Sci Sports Exerc.* 1991;23(3):371-7.
10. Counsilman JE. *The Science of Swimming.* Englewood Cliffs, NJ: Prentice Hall; 1968.
11. de Koning JJ, Noordhof DA, Uitslag TP, Galiart RE, Dodge C, Foster C. An approach to estimating gross efficiency during high-intensity exercise. *Int J Sports Physiol Perform.* 2013;8(6):682-4.
12. Di Prampero P. The energy cost of human locomotion on land and in water. *Int J Sports Med.* 1986;7(2):55-72.
13. Dunbar CC, Robertson RJ, Baun R, Blandin MF, Metz K, Burdett R, et al. The validity of regulating exercise intensity by ratings of perceived exertion. *Med Sci Sports Exerc.* 1992; 24(1):94-9.
14. Foley R, Lahr MM. The role of “the aquatic” in human evolution: Constraining the aquatic ape hypothesis. *Evolutionary Anthropology: Issues, News, and Reviews.* 2014;23(2):56-9.
15. Halson SL. Monitoring training load to understand fatigue in athletes. *Sports Med.* 2014;44(2):139-47.
16. Hawley J, Burke L. *Peak Performance: Training and Nutritional Strategies for Sport.* Sydney, Australia: Allen & Unwin; 1998.
17. Hellard P, Weber S, Rodriguez F, Mader A, editors. Influence of oxidative and glycolytic capacities on the muscular energy metabolism and performance in 200-m freestyle swimming: a case study. 4th FINA World Aquatics Convention; 2016; Windsor, Canada.
18. Hellard P, Scordia C, Avalos M, Mujika I, Pyne DB. Modelling of optimal training load patterns during the 11 weeks preceding major competition in elite swimmers. *Applied Physiology, Nutrition, and Metabolism.* 2017;42(10):1106-17.
19. Hellard P, Pla R, Rodríguez FA, Simbana D, Pyne DB. Dynamics of the metabolic response during a competitive 100-m freestyle in elite male swimmers. *Int J Sports Physiol Perform.* 2018:1-28.
20. Hellard P, Avalos M, Lacoste L, Barale F, Chatard J-C, Millet GP. Assessing the limitations of the Banister model in monitoring training. *J Sports Sci.* 2006;24(5):509-20.
21. Hellard P, Avalos M, Hausswirth C, Pyne D, Toussaint J-F, Mujika I. Identifying optimal overload and taper in elite swimmers over time. *J Sports Sci Med.* 2013;12(4):668.
22. Hellard P, Rodríguez FA, Dupont C, Pyne DB, Mader A, Weber S. *Simulated Metabolic Responses During High-Intensity Interval Training Based on a Mathematical Model in Elite Swimmers.* NOVA; 2018.
23. Holmer I, Astrand P. Swimming training and maximal oxygen uptake. *J Appl Physiol.* 1972;33(4):510-3.
24. Hopkins WG. Quantification of training in competitive sports. *Sports Med.* 1991;12(3):161-83.
25. Huot-Marchand F, Nesi X, Sidney M, Alberty M, Pelayo P. Swimming: Variations of stroking parameters associated with 200 m competitive performance improvement in top-standard front crawl swimmers. *Sports Biomechanics.* 2005;4(1):89-100.
26. Laughlin M, Roseguini B. Mechanisms for exercise training-induced increases in skeletal muscle blood flow capacity: differences with interval sprint training versus aerobic endurance training. *Journal of Physiology and Pharmacology: An Official Journal of the Polish Physiological Society.* 2008;59(Suppl 7):71.
27. Laursen P. Training for intense exercise performance: High-intensity or high-volume training? *Scand J Med Sci Sports.* 2010;20(s2):1-10.
28. Maglischo EW. *Swimming Fastest.* Champaign, IL: Human Kinetics; 2003.
29. Mauger A, Neuloh J, Castle P. Analysis of pacing strategy selection in elite 400-m freestyle swimming. *Med Sci Sports Exerc.* 2012;44(11):2205-12.
30. McGibbon K, Pyne D, Shephard M, Thompson K. Pacing in swimming: A systematic review. Submitted for peer review. 2018.

31. Mujika I. *Tapering and Peaking for Optimal Performance*. Champaign, IL: Human Kinetics 1; 2009.
32. Mujika I, Chatard J-C, Busso T, Geyssant A, Barale F, Lacoste L. Effects of training on performance in competitive swimming. *Can J Appl Physiol*. 1995;20(4):395-406.
33. Olbrecht J. *The Science of Winning: Planning, Periodizing and Optimizing Swim Training*. Tienen, Belgium: F&G Partners; 2015.
34. Paton CD, Hopkins WG. Effects of high-intensity training on performance and physiology of endurance athletes. *Sportscience*. 2004;8:25-41.
35. Pyne D, Touretski G. An analysis of the training of Olympic sprint champion Alexandre Popov. *Australian Swim Coach*. 1993;10(5):5-14.
36. Pyne D, Maw G, Goldsmith W. Protocols for the physiological assessment of swimmers. In: *Physiological Tests for Elite Athletes Australian Sports Commission*. Champaign, IL: Human Kinetics. 2000:372-82.
37. Pyne DB, Sharp RL. Physical and energy requirements of competitive swimming events. *International Journal of Sport Nutrition and Exercise Metabolism*. 2014;24(4):351-9.
38. Robertson E, Pyne D, Hopkins W, Anson J. Analysis of lap times in international swimming competitions. *J Sports Sci*. 2009;27(4):387-95.
39. Rodríguez FA, Mader A. Energy systems in swimming. In: *World Book of Swimming From Science to Performance*. New York: Nova. 2011:225-40.
40. Rushall BS. Swimming energy training in the 21st century: The justification for radical changes. *Swimming Science Bulletin* 2014;(39).
41. Saw AE, Main LC, Gastin PB. Monitoring the athlete training response: Subjective self-reported measures trump commonly used objective measures: A systematic review. *Br J Sports Med*. 2015;bjsports-2015-094758.
42. Secher NH, Volianitis S. Are the arms and legs in competition for cardiac output? *Med Sci Sports Exerc*. 2006;38(10):1797-803.
43. Seiler S, Tønnessen E. SPORTSCIENCE- sportsci.org. *Sportscience*. 2009;13:32-53.
44. Sousa A, Figueiredo P, Oliveira N, Oliveira J, Silva A, Keskinen K, et al. VO2 kinetics in 200-m race-pace front crawl swimming. *Int J Sports Med*. 2011;32(10):765-70.
45. Toussaint HM, Hollander AP. Energetics of competitive swimming. *Sports Med*. 1994;18(6):384-405.

Chapter 18

1. Balsalobre-Fernandez C, Glaister M, Lockett RA. The validity and reliability of an iPhone app for measuring vertical jump performance. *J Sports Sci*. 2015;33(15):1574-9.
2. Baquet G, Berthoin S, Dupont G, Blondel N, Fabre C, Van Praagh E. Effects of high intensity intermittent training on peak $\dot{V}O_2$ in prepubertal children. *Int J Sports Med*. 2002;23(6):439-44.
3. Bosco C, Mognoni P, Luhtanen P. Relationship between isokinetic performance and ballistic movement. *European Journal of Applied Physiology and Occupational Physiology*. 1983;51(3):357-64.
4. Buchheit M. The 30-15 intermittent fitness test: Accuracy for individualizing interval training of young intermittent sport players. *Journal of Strength & Conditioning Research*. 2008;22(2):365-74.
5. Buchheit M, Laursen PB. High-intensity interval training, solutions to the programming puzzle: Part I: Cardiopulmonary emphasis. *Sports Med*. 2013;43(5):313-38.
6. Buchheit M, Morgan W, Wallace J, Bode M, Poulos N. Physiological, psychometric, and performance effects of the Christmas break in Australian football. *Int J Sports Physiol Perform*. 2015;10(1):120-3.
7. Buchheit M, Laursen P, Kuhnle J, Ruch D, Renaud C, Ahmaidi S. Game-based training in young elite handball players. *Int J Sports Med*. 2009;30(4):251.
8. Christensen PM, Krstrup P, Gunnarsson TP, Kiilerich K, Nybo L, Bangsbo J. VO2 kinetics and performance in soccer players after intense training and inactivity. *Medicine and Science in Sports and Exercise*. 2011;43(9):1716-24.
9. Coutts AJ, Gomes RV, Viveiros L, Aoki MS. Monitoring training loads in elite tennis. *Revista Brasileira de Cineantropometria & Desempenho Humano*. 2010;12(3):217-20.
10. DiFiori JP, Benjamin HJ, Brenner JS, Gregory A, Jayanthi N, Landry GL, et al. Overuse injuries and burnout in youth sports: A position statement from the American Medical Society for Sports Medicine. *Br J Sports Med*. 2014;48(4):287-8.

11. Duffield R, Murphy A, Kellett A, Reid M. Recovery from repeated on-court tennis sessions: Combining cold water immersion, compression and sleep recovery interventions. *Int J Sports Physiol Perform*. 2014;9(2):273-82.
12. Fernandez-Fernandez J, Sanz-Rivas D, Mendez-Villanueva A. A review of the activity profile and physiological demands of tennis match play. *Strength Cond J*. 2009;31(4):15-26.
13. Fernandez-Fernandez J, Ulbricht A, Ferrauti A. Fitness testing of tennis players: How valuable is it? *Br J Sports Med*. 2014;48(Suppl 1):i22-31.
14. Fernandez-Fernandez J, Sanz-Rivas D, Fernandez-Garcia B, Mendez-Villanueva A. Match activity and physiological load during a clay-court tennis tournament in elite female players. *J Sports Sci*. 2008;26(14):1589-95.
15. Fernandez-Fernandez J, Zimek R, Wiewelhove T, Ferrauti A. High-intensity interval training vs. repeated-sprint training in tennis. *J Strength Cond Res*. 2012;26(1):53-62.
16. Fernandez-Fernandez J, Sanz-Rivas D, Kovacs MS, Moya M. In-season effect of a combined repeated sprint and explosive strength training program on elite junior tennis players. *J Strength Cond Res*. 2015;29(2):351-7.
17. Fernandez-Fernandez J, Sanz-Rivas D, Sarabia JM, Moya M. Preseason training: The effects of a 17-day high-intensity shock microcycle in elite tennis players. *J Sports Sci Med*. 2015;14(4):783-91.
18. Fernandez-Fernandez J, Sanz D, Sarabia JM, Moya M. The effects of sport-specific drills training or high-intensity interval training in young tennis players. *Int J Sports Physiol Perform*. 2017;12(1):90-8.
19. Fernandez-Fernandez J, Sanz-Rivas D, Sanchez-Munoz C, de la Aleja Tellez JG, Buchheit M, Mendez-Villanueva A. Physiological responses to on-court vs running interval training in competitive tennis players. *J Sports Sci Med*. 2011;10(3):540-5.
20. Fernandez-Fernandez J, Boullousa DA, Sanz-Rivas D, Abreu L, Filaire E, Mendez-Villanueva A. Psychophysiological stress responses during training and competition in young female competitive tennis players. *Int J Sports Med*. 2015;36(1):22-8.
21. Fernandez J, Mendez-Villanueva A, Pluim B. Intensity of tennis match play. *Br J Sports Med*. 2006;40(5):387-91.
22. Ferrauti A, Kinner V, Fernandez-Fernandez J. The Hit & Turn Tennis Test: An acoustically controlled endurance test for tennis players. *J Sports Sci*. 2011;29(5):485-94.
23. Foster C, Florhaug JA, Franklin J, Gottschall L, Hrovatin LA, Parker S, et al. A new approach to monitoring exercise training. *J Strength Cond Res*. 2001;15(1):109-15.
24. Gallo-Salazar C, Del Coso J, Barbado D, Lopez-Valenciano A, Santos-Rosa FJ, Sanz-Rivas D, et al. Impact of a competition with two consecutive matches in a day on physical performance in young tennis players. *Appl Physiol Nutr Metab*. 2017;42(7):750-6.
25. Gibala MJ, McGee SL. Metabolic adaptations to short-term high-intensity interval training: A little pain for a lot of gain? *Exerc Sport Sci Rev*. 2008;36(2):58-63.
26. Girard O, Millet GP. Physical determinants of tennis performance in competitive teenage players. *J Strength Cond Res*. 2009;23(6):1867-72.
27. Girard O, Lattier G, Maffiuletti NA, Micallef J-P, Millet GP. Neuromuscular fatigue during a prolonged intermittent exercise: Application to tennis. *Journal of Electromyography and Kinesiology*. 2008;18(6):1038-46.
28. Gomes RV, Moreira A, Lodo L, Nosaka K, Coutts AJ, Aoki MS. Monitoring training loads, stress, immune-endocrine responses and performance in tennis players. *Biol Sport*. 2013;30(3):173-80.
29. Harrison CB, Gill ND, Kinugasa T, Kilding AE. Development of aerobic fitness in young team sport athletes. *Sports Med*. 2015;45(7):969-83.
30. Haydar B, Al Haddad H, Ahmaidi S, Buchheit M. Assessing inter-effort recovery and change of direction ability with the 30-15 Intermittent Fitness Test. *J Sports Sci Med*. 2011;10(2):346.
31. Hill-Haas S, Coutts A, Rowsell G, Dawson B. Generic versus small-sided game training in soccer. *Int J Sports Med*. 2009;30(9):636-42.
32. Jayanthi N, Feller E, Smith A. Junior competitive tennis: Ideal training and tournament recommendations. *J Med Sci Tennis*. 2013;18:30-6.

33. Kovacs MS. Tennis physiology: Training the competitive athlete. *Sports Med.* 2007;37(3):189-98.
34. Laursen PB, Jenkins DG. The scientific basis for high-intensity interval training: Optimising training programmes and maximising performance in highly trained endurance athletes. *Sports Med.* 2002;32(1):53-73.
35. Mendez-Villanueva A, Fernandez-Fernandez J, Bishop D. Exercise-induced homeostatic perturbations provoked by singles tennis match play with reference to development of fatigue. *Br J Sports Med.* 2007;41(11):717-22; discussion 22.
36. Mendez-Villanueva A, Fernandez-Fernandez J, Bishop D, Fernandez-Garcia B, Terrados N. Activity patterns, blood lactate concentrations and ratings of perceived exertion during a professional singles tennis tournament. *Br J Sports Med.* 2007;41(5):296-300.
37. Moreno-Pérez V, Ayala F, Fernandez-Fernandez J, Vera-Garcia FJ. Descriptive profile of hip range of motion in elite tennis players. *Physical Therapy in Sport.* 2016;19:43-8.
38. Murphy A, Duffield R, Kellett A, Reid M. The relationship of training load to physical capacity changes during international tours in high performance junior tennis players. *Int J Sports Physiol Perform.* 2015;10(2):253-60.
39. Murphy A, Duffield R, Kellett A, Reid M. A descriptive analysis of internal and external loads for elite-level tennis drills. *Int J Sports Physiol Perform.* 2014;9(5):863-70.
40. Murphy AP, Duffield R, Kellett A, Reid M. A descriptive analysis of internal and external loads for elite-level tennis drills. *Int J Sports Physiol Perform.* 2014;9(5):863-70.
41. Pluim BM, Staal JB, Windler GE, Jayanthi N. Tennis injuries: Occurrence, aetiology, and prevention. *Br J Sports Med.* 2006;40(5):415-23.
42. Reid M, Duffield R, Dawson B, Baker J, Crespo M. Quantification of the physiological and performance characteristics of on-court tennis drills. *Br J Sports Med.* 2008;42(2):146-51.
43. Reid MM, Duffield R, Minett GM, Sibte N, Murphy AP, Baker J. Physiological, perceptual, and technical responses to on-court tennis training on hard and clay courts. *J Strength Cond Res.* 2013;27(6):1487-95.
44. Smekal G, von Duvillard SP, Rihacek C, Pokan R, Hofmann P, Baron R, et al. A physiological profile of tennis match play. *Med Sci Sports Exerc.* 2001;33(6):999-1005.
45. Sperlich B, De Marées M, Koehler K, Linville J, Holmberg H-C, Mester J. Effects of 5 weeks' high-intensity interval training vs. volume training in 14-year-old soccer players. *J Strength Cond Res.* 2011;25(5):1271-8.
46. Ulbricht A, Fernandez-Fernandez J, Mendez-Villanueva A, Ferrauti A. Impact of fitness characteristics on tennis performance in elite junior tennis players. *J Strength Cond Res.* 2016;30(4):989-98.
47. Wiewelhove T, Fernandez-Fernandez J, Raeder C, Kappenstein J, Meyer T, Kellmann M, et al. Acute responses and muscle damage in different high-intensity interval running protocols. *J Sports Med Phys Fitness.* 2016;56(5):606-15.

Chapter 19

1. Abbiss CR, Laursen PB. Describing and understanding pacing strategies during athletic competition. *Sports Medicine.* 2008;38(3):239-52.
2. Allen H, Coggan A. *Training and Racing With a Power Meter.* Boulder, CO: VeloPress; 2010.
3. Banister EW. Modeling elite athletic performance. In: MacDougall JD, Wenger HA, Green HJ, eds. *Physiological Testing of Elite Athletes.* Champaign, IL: Human Kinetics; 1991.
4. Banister EW, Calvert TW. Planning for future performance: Implications for long-term training. *Can J Appl Sport Sci.* 1980;5(3):170-6.
5. Barnes KR, Kilding AE. Strategies to improve running economy. *Sports Med.* 2015;45(1):37-56.
6. Bentley DJ, Millet GP, Vleck VE, McNaughton LR. Specific aspects of contemporary triathlon: Implications for physiological analysis and performance. *Sports Med.* 2002;32(6):345-59.
7. Bentley DJ, McNaughton LR, Lamymann R, Roberts SP. The effects of prior incremental cycle exercise on the physiological responses during incremental running to exhaustion: Relevance for sprint triathlon performance. *J Sports Sci.* 2003;21(1):29-38.

8. Bentley DJ, Cox GR, Green D, Laursen PB. Maximising performance in triathlon: applied physiological and nutritional aspects of elite and non-elite competitions. *J Sci and Med Sport / Sports Med Australia*. 2008;11(4):407-16.
9. Burke LM. Fueling strategies to optimize performance: Training high or training low? *Scandinavian J Med Sci Sports*. 2010;20 (Suppl 2):48-58.
10. Burke LM, Hawley JA, Wong SH, Jeukendrup AE. Carbohydrates for training and competition. *J Sports Sci*. 2011;29 (Suppl 1):S17-27.
11. Frandsen J, Vest SD, Larsen S, Dela F, Helge JW. Maximal fat oxidation is related to performance in an Ironman triathlon. *Int J Sports Med*. 2017;38(13):975-82.
12. Hausswirth C, Bigard AX, Guezennec CY. Relationships between running mechanics and energy cost of running at the end of a triathlon and a marathon. *Int J Sports Med*. 1997;18(5):330-9.
13. Hooper SL, Mackinnon LT, Howard A, Gordon RD, Bachmann AW. Markers for monitoring overtraining and recovery. *Med Sci Sports Exerc*. 1995;27(1):106-12.
14. Issurin VB. New horizons for the methodology and physiology of training periodization. *Sports Med*. 2010;40(3):189-206.
15. Laursen PB. Long distance triathlon: Demands, preparation and performance. *J Hum Sport Exerc*. 2011;6(2):247-63.
16. Laursen PB, Rhodes EC. Factors affecting performance in an ultraendurance triathlon. *Sports Med*. 2001;31(3):195-209.
17. Laursen PB, Rhodes EC, Langill RH, McKenzie DC, Taunton JE. The effect of the cycle phase on run performance in an ultraendurance triathlon. *Adv Exerc Sports Physiol*. 2001;7(3):81-6.
18. Laursen PB, Knez WL, Shing CM, Langill RH, Rhodes EC, Jenkins DG. Relationship between laboratory-measured variables and heart rate during an ultra-endurance triathlon. *Journal of Sports Sciences*. 2005;23(10):1111-20.
19. Laursen PB, Knez WL, Shing CM, Langill RH, Rhodes EC, Jenkins DG. Relationship between laboratory-measured variables and heart rate during an ultra-endurance triathlon. *J Sports Sci*. 2005;23(10):1111-20.
20. Marquet LA, Brisswalter J, Louis J, Tiollier E, Burke LM, Hawley JA, et al. Enhanced endurance performance by periodization of carbohydrate intake: "Sleep low" strategy. *Med Sci Sports Exerc*. 2016;48(4):663-72.
21. Morton RH, Fitz-Clarke JR, Banister EW. Modeling human performance in running. *J Appl Physiol* (1985). 1990;69(3):1171-7.
22. Mujika I. Olympic preparation of a world-class female triathlete. *Int J Sports Physiol Perform*. 2014;9(4):727-31.
23. Nimmerichter A, Eston R, Bachl N, Williams C. Effects of low and high cadence interval training on power output in flat and uphill cycling time-trials. *Eur J Appl Physiol*. 2012;112(1):69-78.
24. Plews DJ, Laursen PB, Kilding AE, Buchheit M. Heart rate variability in elite triathletes, is variation in variability the key to effective training? A case comparison. *Eur J Appl Physiol*. 2012;112(11):3729-41.
25. Plews DJ, Laursen PB, Stanley J, Kilding AE, Buchheit M. Training adaptation and heart rate variability in elite endurance athletes: Opening the door to effective monitoring. *Sports Med*. 2013;43(9):773-81.
26. Plews DJ, Scott B, Altini M, Wood M, Kilding AE, Laursen PB. Comparison of heart-rate-variability recording with smartphone photoplethysmographic, Polar H7 chest strap, and electrocardiogram methods. *Int J Sports Physiol Perform*. 2017;1-17.
27. Sedgwick PE, Wortley GC, Wright JM, Asplund C, Roberts WO, Usman S. Medical clearance for desert and land sports, adventure, and endurance events. *Clin J Sport Med*. 2015;25(5):418-24.
28. Telles T, Barroso R, Figueiredo P, Salgueiro DF, Vilas-Boas JP, Junior OA. Effect of hand paddles and parachute on backstroke coordination and stroke parameters. *J Sports Sci*. 2017;35(9):906-11.
29. Volek JS, Freidenreich DJ, Saenz C, Kunces LJ, Creighton BC, Bartley JM, et al. Metabolic characteristics of keto-adapted ultra-endurance runners. *Metabolism*. 2016;65(3):100-10.
30. Wakayoshi K, Ikuta K, Yoshida T, Udo M, Moritani T, Mutoh Y, et al. Determination and validity of critical velocity as an index of swimming performance in the competitive swimmer. *Eur J Appl Physiol Occup Physiol*. 1992;64(2):153-7.
31. Webster CC, Swart J, Noakes TD, Smith JA. A carbohydrate ingestion intervention in an elite athlete who follows a LCHF diet. *Int J Sports Physiol Perform*. 2017;1-12.

32. Wood T. Lost metabolic machinery during ketosis? Depends where you are looking. *Strength Cond J*. 2017;39(5):94-5.
33. Yeo WK, Lessard SJ, Chen ZP, Garnham AP, Burke LM, Rivas DA, et al. Fat adaptation followed by carbohydrate restoration increases AMPK activity in skeletal muscle from trained humans. *J Appl Physiol*. 2008;105(5):1519-26.

Chapter 20

1. Bosch TA, Burruss TP, Weir NL, Fielding KA, Engel BE, Weston TD, et al. Abdominal body composition differences in NFL football players. *J Strength Cond Res*. 2014;28(12):3313-9.
2. Brophy RH, Wright RW, Powell JW, Matava MJ. Injuries to kickers in American football: The National Football League experience. *Am J Sports Med*. 2010;38(6):1166-73.
3. Brophy RH, Lyman S, Chehab EL, Barnes RP, Rodeo SA, Warren RF. Predictive value of prior injury on career in professional American football is affected by player position. *Am J Sports Med*. 2009;37(4):768-75.
4. Buchheit M, Laursen PB. High-intensity interval training, solutions to the programming puzzle. Part II: Anaerobic energy, neuromuscular load and practical applications. *Sports Med*. 2013;43(10):927-54.
5. Buchheit M, Laursen PB. High-intensity interval training, solutions to the programming puzzle: Part I: Cardiopulmonary emphasis. *Sports Med*. 2013;43(5):313-38.
6. Cormack SJ, Newton RU, McGuigan MR. Neuromuscular and endocrine responses of elite players to an Australian rules football match. *Int J Sports Physiol Perform*. 2008;3(3):359-74.
7. Dyreson M. The Super Bowl as a television spectacle: Global designs, global niches, and parochial patterns. *International Journal of the History of Sport*. 2017;34(1-2):139-56.
8. Edwards T, Spiteri T, Piggott B, Haff GG, Joyce C. A narrative review of the physical demands and injury incidence in American football: Application of current knowledge and practices in workload management. *Sports Med*. 2017:1-11.
9. Fry AC, Kraemer WJ. Physical performance characteristics of American collegiate football players. *J Strength Cond Res*. 1991;5(3):126-38.
10. Fullagar HH, McCunn R, Murray A. An updated review of the applied physiology of American collegiate football: The physical demands, strength/conditioning, nutritional considerations and injury characteristics of America's favourite game. *Int J Sports Physiol Perform*. 2017;12(10):1396-1403.
11. Gabbett TJ. The training–injury prevention paradox: Should athletes be training smarter and harder? *Br J Sports Med*. 2016;50(5):273-80.
12. Gallo TF, Cormack SJ, Gabbett TJ, Lorenzen CH. Pre-training perceived wellness impacts training output in Australian football players. *J Sports Sci*. 2016;34(15):1445-51.
13. Govus AD, Coutts A, Duffield R, Murray A, Fullagar H. Relationship between pre-training subjective wellness measures, player load and rating of perceived exertion training load in American college football. *Int J Sports Physiol Perform*. 2018;13(1):95-101.
14. Hoffman J. Physiological demands of American football. *Sports Science Exchange*. 2015;28(143):1-6.
15. Hoffman JR. The applied physiology of American football. *Int J Sports Physiol Perform*. 2008;3(3):387-92.
16. Hoffman JR, Kang J, Ratamess NA, Faigenbaum AD. Biochemical and hormonal responses during an intercollegiate football season. *Med Sci Sports Exerc*. 2005;37(7):1237-41.
17. Hoffman JR, Maresh CM, Newton RU, Rubin MR, French DN, Volek JS, et al. Performance, biochemical, and endocrine changes during a competitive football game. *Med Sci Sports Exerc*. 2002;34(11):1845-53.
18. Hopsicker P, Dyreson M. Super Bowl Sunday: A national holiday and a global curiosity. *International Journal of the History of Sport*. 2017;34(1-2):1-6.
19. Impellizzeri FM, Rampinini E, Coutts AJ, Sassi A, Marcora SM. Use of RPE-based training load in soccer. *Med Sci Sports Exerc*. 2004;36(6):1042-7.
20. Iosia MF, Bishop PA. Analysis of exercise-to-rest ratios during division IA televised football competition. *J Strength Cond Res*. 2008;22(2):332-40.
21. Jacobson BH, Conchola EG, Glass RG, Thompson BJ. Longitudinal morphological and performance profiles for American, NCAA Division I

- football players. *J Strength Cond Res.* 2013;27(9):2347-54.
22. Kraemer W, Gotshalk L. *Physiology of American Football*. Baltimore, MD: Lippincott, Williams & Wilkins; 2000:795-813.
 23. Kraemer WJ, Torine JC, Dudley J, Martin GJ. Nonlinear periodization: Insights for use in collegiate and professional American football resistance training programs. *Strength Cond J.* 2015;37(6):17-36.
 24. Kraemer WJ, Torine JC, Silvestre R, French DN, Ratamess NA, Spiering BA, et al. Body size and composition of National Football League players. *J Strength Cond Res.* 2005;19(3):485-9.
 25. MacLean SA, Basch CH, Garcia P. Alcohol and violence in 2017 National Football League Super Bowl commercials. *Health Promotion Perspectives.* 2017;7(3):163-7.
 26. Malone S, Roe M, Doran DA, Gabbett TJ, Collins K. High chronic training loads and exposure to bouts of maximal velocity running reduce injury risk in elite Gaelic football. *J Sci Med Sport.* 2017;20(3):250-4.
 27. Malone S, Owen A, Newton M, Mendes B, Tiernan L, Hughes B, et al. Wellbeing perception and the impact on external training output among elite soccer players. *J Sci Med Sport.* 2018;21(1):29-34.
 28. McCall A, Arden C, Delecroix B, Abaidia A-e, Dunlop G, Dupont G. Adding a quick and simple psychological measure of player readiness into the return to play mix: A single player case study from professional football (soccer). *Sport Performance & Science Reports.* Nov. 2017;1:1-3. https://sportperfsci.com/wp-content/uploads/2017/11/SPSR8_McCall-et-al._171110_8v1_final.pdf.
 29. National Football League Seasonal Statistics 2017. Available from <http://www.nfl.com/stats/>.
 30. Pincivero DM, Bompa TO. A physiological review of American football. *Sports Med.* 1997;23(4):247-60.
 31. Pryor JL, Huggins RA, Casa DJ, Palmieri GA, Kraemer WJ, Maresh CM. A Profile of a National Football League team. *J Strength Cond Res.* 2014;28(1):7-13.
 32. Robbins DW. The National Football League (NFL) Combine: Does normalized data better predict performance in the NFL draft? *J Strength Cond Res.* 2010;24(11):2888-99.
 33. Robbins DW. Positional physical characteristics of players drafted into the National Football League. *J Strength Cond Res.* 2011;25(10):2661-7.
 34. Rogalski B, Dawson B, Heasman J, Gabbett TJ. Training and game loads and injury risk in elite Australian footballers. *J Sci Med Sport.* 2013;16(6):499-503.
 35. Sierer SP, Battaglini CL, Mihalik JP, Shields EW, Tomasini NT. The National Football League Combine: Performance differences between drafted and nondrafted players entering the 2004 and 2005 drafts. *J Strength Cond Res.* 2008;22(1):6-12.
 36. Stone J, Kreutzer A, Mata J, Nystrom M, Jagim A, Jones M, et al. Changes in creatine kinase and hormones over the course of an American football season. *J Strength Cond Res.* 2017.
 37. Trexler ET, Smith-Ryan AE, Mann JB, Ivey PA, Hirsch KR, Mock MG. Longitudinal body composition changes in NCAA Division I college football players. *J Strength Cond Res.* 2017;31(1):1-8.
 38. Ward PA, Ramsden S, Coutts AJ, Hulton AT, Drust B. Positional differences in running and non-running activities during elite American football training. *J Strength Cond Res.* 2018;32(7):2072-84.
 39. Wellman AD, Coad SC, Goulet GC, McLellan CP. Quantification of accelerometer derived impacts associated with competitive games in National Collegiate Athletic Association Division I college football players. *J Strength Cond Res.* 2017;31(2):330-8.
 40. Wellman AD, Coad SC, Flynn PJ, Climstein M, McLellan CP. Movement demands and perceived wellness associated with preseason training camp in NCAA Division I college football players. *J Strength Cond Res.* 2017;31(10):2704-18.
 41. Wellman AD, Coad SC, Flynn PJ, Siam TK, McLellan CP. A comparison of pre-season and in-season practice and game loads in NCAA Division I football players. *J Strength Cond Res.* Jul 2017.
 42. Yamashita D, Asakura M, Ito Y, Yamada S, Yamada Y. Physical characteristics and perfor-

mance of Japanese top-level American football players. *J Strength Cond Res.* 2017;31(9):2455-61.

Chapter 21

1. Aughey RJ. Widening margin in activity profile between elite- and sub-elite Australian football: A case study. *J Sci Med Sport.* 2013;16(4):382-6.
2. Austin DJ, Gabbett TJ, Jenkins DJ. Repeated high-intensity exercise in a professional rugby league. *J Strength Cond Res.* 2011;25(7):1898-904.
3. Bilsborough JC, Greenway K, Opar D, Livingstone S, Cordy J, Coutts AJ. Changes in body composition during the competition season: Relationships with training, physical performance and team selection in Australian Football players. *Int J Sports Physiol Perf.* 2014;In review.
4. Bilsborough JC, Greenway K, Opar D, Livingstone S, Cordy JT, Bird S, et al. Comparison of anthropometry, upper body strength and lower body power characteristics in different levels of Australian Football players. *J Strength Cond Res.* 2014;29(3):826-34.
5. Buchheit M. Monitoring training status with HR measures: Do all roads lead to Rome? *Frontiers in Physiology.* 2014;5(73).
6. Buchheit M, Laursen PB. High-intensity interval training, solutions to the programming puzzle: Part I: Cardiopulmonary emphasis. *Sports Med.* 2013;43(5):313-38.
7. Buchheit M, Laursen PB. High-intensity interval training, solutions to the programming puzzle. Part II: Anaerobic energy, neuromuscular load and practical applications. *Sports Med.* 2013;43(10):927-54.
8. Buchheit M, Racinais S, Bilsborough JC, Bourdon PC, Voss SC, Hocking J, et al. Monitoring fitness, fatigue and running performance during a pre-season training camp in elite football players. *J Sci Med Sport.* 2013;16(6):550-5.
9. Burgess D, Naughton G, Norton K. Quantifying the gap between under 18 and senior AFL football: 2003-2009. *Int J Sports Physiol Perf.* 2012;7(1):53-8.
10. Corbett DM, Sweeting AJ, Robertson S. Weak relationships between stint duration, physical and skilled match performance in Australian Football. *Frontiers in Physiology.* 2017;8:820.
11. Corbett DM, Bartlett JD, O'Connor F, Back N, Torres-Ronda L, Robertson S. Development of physical and skill training drill prescription systems for elite Australian Rules football. *Science and Medicine in Football.* 2017:1-7.
12. Coutts AJ. In the age of technology, Occam's razor still applies. *Int J Sports Physiol Perf.* 2014;9(5):741.
13. Coutts AJ. Challenges in developing evidence-based practice in high-performance sport. *Int J Sports Physiol Perf.* 2017;12(6):717-8.
14. Coutts AJ, Crowcroft S, Kempton T. Developing athlete monitoring systems: Theoretical basis and practical applications. In: Kellmann M, Beckmann J, eds. *Sport, Recovery and Performance: Interdisciplinary Insights.* Abingdon: Routledge; 2018.
15. Coutts AJ, Quinn J, Hocking J, Castagna C, Rampinini E. Match running performance in elite Australian Rules Football. *J Sci Med Sport.* 2010;13(5):543-8.
16. Delaney JA, Thornton HR, Burgess DJ, Dascombe BJ, Duthie GM. Duration-specific running intensities of Australian Football match-play. *J Sci Med Sport.* 2017;20(7):689-94.
17. Duhig S, Shield AJ, Opar D, Gabbett TJ, Ferguson C, Williams M. Effect of high-speed running on hamstring strain injury risk. *Brit J Sport Med.* 2016;50(24):1536-40.
18. Farrow D, Pyne D, Gabbett T. Skill and physiological demands of open and closed training drills in Australian Football. *International Journal of Sports Science & Coaching.* 2008;3(4):485-94.
19. Foster C, Florhaug JA, Franklin J, Gottschall L, Hrovatin LA, Parker S, et al. A new approach to monitoring exercise training. *J Strength Cond Res.* 2001;15(1):109-15.
20. Gabbett TJ. The training-injury prevention paradox: Should athletes be training smarter and harder? *Br J Sports Med.* 2016;50(5):273-80.
21. Gray AJ, Jenkins DG. Match analysis and the physiological demands of Australian football. *Sports Med.* 2010;40(4):347-60.
22. Hill-Haas SV, Dawson B, Impellizzeri FM, Coutts AJ. Physiology of small-sided games training in football: A systematic review. *Sports Med.* 2011;41(3):199-220.

23. Hiscock D, Dawson B, Heasman J, Peeling P. Game movements and player performance in the Australian Football League. *International Journal of Performance Analysis in Sport*. 2017;12(3):531-45.
24. Johnston RJ, Watsford ML, Pine MJ, Spurrs RW, Murphy A, Pruyn EC. Movement demands and match performance in professional Australian football. *Int J Sport Med*. 2012;33(2):89-93.
25. Kempton T, Sirotic AC, Coutts AJ. Metabolic power demands of rugby league match-play. *Int J Sports Physiol Perf*. 2015;10(1):23-8.
26. Kempton T, Sullivan CJ, Bilsborough JC, Cordy J, Coutts AJ. Match-to-match variation in physical activity and technical skill measures in professional Australian Football. *J Sci Med Sport*. 2015;18(1):109-13.
27. Loader J, Montgomery PG, Williams MD, Lorenzen C, Kemp JG. Classifying training drills based on movement demands in Australian Football. *Int J Sports Sci Coach*. 2012;7(1):57-68.
28. Malone S, Owen A, Mendes B, Hughes B, Collins K, Gabbett TJ. High-speed running and sprinting as an injury risk factor in soccer: Can well-developed physical qualities reduce the risk? *J Sci Med Sport*. 2018;21(3):257-62.
29. McLean BD, Coutts AJ, Kelly V, McGuigan MR, Cormack SJ. Neuromuscular, endocrine, and perceptual fatigue responses during different length between-match microcycles in professional rugby league players. *Int J Sports Physiol Perf*. 2010;5(3):367-83.
30. Moreira A, Kempton T, Aoki MS, Sirotic AC, Coutts AJ. The impact of three different length between-match microcycles on training loads in professional rugby league players. *Int J Sports Physiol Perf*. 2015;10(6):767-73.
31. Murray NB, Gabbett TJ, Townshend AD. Relationship between pre-season training load and in-season availability in elite Australian football players. *Int J Sports Physiol Perf*. 2017;12(6):749-55.
32. Racinais S, Buchheit M, Bilsborough J, Bourdon PC, Cordy J, Coutts AJ. Physiological and performance responses to a training camp in the heat in professional Australian football players. *Int J Sports Physiol Perf*. 2014;9(4):598-603.
33. Rampinini E, Impellizzeri F, Castagna C, Abt G, Chamari K, Sassi A, et al. Factors influencing physiological responses to small-sided soccer games. *J Sports Sci*. 2007;6(25):659-66.
34. Ritchie D, Hopkins WG, Buchheit M, Cordy J, Bartlett JD. Quantification of training and competition load across a season in an elite Australian Football club. *Int J Sports Physiol Perf*. 2016;11(4):474-9.
35. Ryan AS, Coutts AJ, Hocking J, Dillon P, Kempton TK. Factors affecting match running performance in professional Australian Football. *Int J Sports Physiol Perf*. 2017;12(9):1199-204.
36. Scott TJ, Black CR, Quinn J, Coutts AJ. Validity and reliability of the session-RPE method for quantifying training in Australian football: A comparison of the CR10 and CR100 scales. *J Strength Cond Res*. 2013;27(1):270-6.
37. Slattery KM, Wallace LK, Bentley DJ, Coutts AJ. Effect of training load on simulated team sport match performance. *Appl Physiol Nutr Metab*. 2012;37(2):315-22.
38. Spencer M, Lawrence S, Rechichi C, Bishop D, Dawson B, Goodman C. Time-motion analysis of elite field hockey, with special reference to repeated-sprint activity. *J Sports Sci*. 2004;22(9):843-50.
39. Sullivan C, Bilsborough JC, Cianciosi M, Hocking J, Cordy JT, Coutts AJ. Factors affecting match performance in professional Australian football. *Int J Sports Physiol Perf*. 2014;9(3):561-6.
40. Sullivan C, Bilsborough JC, Cianciosi M, Hocking J, Cordy J, Coutts AJ. Match score affects activity profile and skill performance in professional Australian Football players. *J Sci Med Sport*. 2014;17:326-31.
41. Varley MC, Gabbett T, Aughey RJ. Activity profiles of professional soccer, rugby league and Australian football match play. *J Sports Sci*. 2014;32(20):1858-66.
42. Young WB, Newton RU, Doyle TL, Chapman D, Cormack S, Stewart G, et al. Physiological and anthropometric characteristics of starters and non-starters and playing positions in elite Australian Rules Football: a case study. *J Sci Med Sport*. 2005;8(3):333-45.

Chapter 22

1. Camp CL, Zajac JM, Pearson DB, Sinatro AM, Spiker AM, Werner BC, et al. Decreased shoulder external rotation and flexion are greater predictors of injury than internal rotation deficits:

- Analysis of 132 pitcher-seasons in professional baseball. *Arthroscopy*. 2017;33(9):1629-36.
2. Cornell, DJ, Paxson, JL, Caplinger, RA, Seligman, JR, Davis, NA, Flees, RJ, et al. In-game heart rate responses among professional baseball starting pitchers. *J Strength Cond Res*. 2017;31(1):24-9.
 3. Gibala M, McGee S. Metabolic adaptations to short-term high-intensity interval training: A little pain for a lot of gain? *Exerc & Sports Sci Reviews*. 2008;36(2):58-63.
 4. Gillett J, Dawes J, Spaniol F, Rhea M, Rogowski J, Magrini M, et al. A description and comparison of cardiorespiratory fitness measures in relation to pitching performance among professional baseball pitchers. *J Strength Cond Res*. 2016;4(14):1-8.
 5. Hoffman J, Vazquez J, Pichardo N, and Tenenbaum G. Anthropometric and performance comparisons in professional baseball players. *J Strength Cond Res*. 2009;23(8):2173-8.
 6. Hughes L, Paton B, Rosenblatt B, Gissane C, Patterson SD. Blood flow restriction training in clinical musculoskeletal rehabilitation: A systematic review and meta-analysis. *Br J Sports Med*. 2107;51(13):1003-11.
 7. Jones CM, Griffiths PC, Mellalieu SD. Training load and fatigue marker associations with injury and illness: A systematic review of longitudinal studies. *Sports Med*. 2017;47(5):943-74.
 8. Mangan G, Hoffman J, Fragala M, Vazquez J, Krause M, Gillett J, et al. Effect of age on anthropometric and physical performance measures in professional baseball players. *J Strength Cond Res*. 2013;27(2):375-81.
 9. Mangan G, Hoffman J, Vazquez J, Pichardo N, Fragala M, and Stout J. Predictors of fielding performance in professional baseball players. *Int J Sports Physiol Perf*. 2013;8:510-16.
 10. Rhea M, Oliverson J, Marshall G, Peterson M, Kenn J, and Naclerio Ayllon F. Noncompatibility of power and endurance training among college baseball players. *J Strength Cond Res*. 2008;22(1):230-4.
 11. Spaniol F. Body composition and baseball performance. *Perf Training J*. 2005;4(1):10-11.
 12. Szymanski DJ. Physiology of baseball pitching dictates specific exercise intensity for conditioning. *Strength Cond J*. 2009;31(2):41-7.
 13. Wilk KE, Macrina LC, Fleisig GS, Aune KT, Porterfield RA, Harker P, et al. Deficits in glenohumeral passive range of motion increase risk of shoulder injury in professional baseball pitchers: A prospective study. *Am J Sports Med*. 2015;43(10):2379-85.
 14. Yang SC, Wang CC, Lee SD, Lee YU, Chan KH, Chen YL, et al. Impact of 12-s rule on performance and muscle damage of baseball pitchers. *Med Sci Sports Exerc*. 2016;48(12):2512-16.

Chapter 23

1. Aglioti SM, Cesari P, Romani M, Urgesi C. Action anticipation and motor resonance in elite basketball players. *Nature Neuroscience*. 2008;11(9):1109-16.
2. Aguiar M, Botelho G, Lago C, Maças V, Sampaio J. A review on the effects of soccer small-sided games. *J Hum Kinet*. 2012;33:103-13.
3. Alexander J, Hopkinson T, Wundersitz D, Serpell B, Mara J, Ball N. Validity of a wearable accelerometer device to measure average acceleration values during high-speed running. *J Strength Cond Res*. 2016;30(11):3007-13.
4. Astorino TA, Allen RP, Roberson DW, Juranich M. Effect of high-intensity interval training on cardiovascular function, VO₂max, and muscular force. *J Strength Cond Res*. 2012;26(1):138-45.
5. Atl H, Köklü Y, Alemdaroğlu U, Koçak FU. A comparison of heart rate response and frequencies of technical actions between half-court and full-court 3-a-side games in high school female basketball players. *J Strength Cond Res*. 2013;27(2):352-6.
6. Bartlett JD, O'Connor F, Pitchford N, Torres-Ronda L, Robertson SJ. Relationships between internal and external training load in team-sport athletes: Evidence for an individualized approach. *Int J Sports Physiol Perform*. 2017;12(2):230-4.
7. Ben Abdelkrim N, El Fazaa S, El Ati J. Time-motion analysis and physiological data of elite under-19-year-old basketball players during competition. *Brit J Sport Med*. 2007;41(2):69-75.
8. Ben Abdelkrim N, Castagna C, Jabri I, Battikh T, El Fazaa S, El Ati J. Activity profile and physiological requirements of junior elite basketball players in relation to aerobic-anaerobic fitness. *J Strength Cond Res*. 2010;24(9):2330-42.

9. Buchheit M. Monitoring training status with HR measures: Do all roads lead to Rome? *Front Physiol.* Feb 2014;27(5):73.
10. Buchheit M, Laursen PB. High-intensity interval training, Solutions to the programming puzzle: Part I: Cardiopulmonary emphasis. *Sports Med.* 2013;43(5):313-38.
11. Burns AS, Lauder TD. Deep water running: an effective non-weightbearing exercise for the maintenance of land-based running performance. *Mil Med.* 2001;166(3):253-8.
12. Cardinale M, Varley MC. Wearable training monitoring technology: Applications, challenges and opportunities. *Int J Sports Physiol Perform.* 2017;12(Suppl 2):S255–S62.
13. Castagna C, Impellizzeri FM, Chaouachi A, Ben Abdelkrim N, Manzi V. Physiological responses to ball-drills in regional level male basketball players. *J Sports Sci.* 2011;29(12):1329-36.
14. Castagna C, Impellizzeri F, Chaouachi A, Ben Abdelkrim N, Manzi V. Physiological responses to ball-drills in regional level male basketball players. *J Sports Sci.* 2011;29(12):1329-36.
15. Castagna C, Manzi V, D'Ottavio S, Annino G, Padua E, Bishop D. Relation between maximal aerobic power and the ability to repeat sprints in young basketball players. *J Strength Cond Res.* 2007;21(4):1172-6.
16. Clark IE, West BM, Reynolds SK, Murray SR, Pettitt RW. Applying the critical velocity model for an off-season interval training program. *J Strength Cond Res.* 2013;27(12):3335-41.
17. Davids K, Araújo D, Correia V, Vilar L. How small-sided and conditioned games enhance acquisition of movement and decision-making skills. *Exerc Sport Sci Rev.* 2013;41(3):154-61.
18. Delextrat A, Kraiem S. Heart-rate responses by playing position during ball drills in basketball. *Int J Sports Physiol Perform.* 2013;8:410-8.
19. Delextrat A, Martinez A. small-sided game training improves aerobic capacity and technical skills in basketball players. *Int J Sports Med.* 2014;35(5):385-91.
20. Dellal A, Chamari K, Pintus A, Girard O, Cotte T, Keller D. Heart rate responses during small-sided games and short intermittent running training in elite soccer players: A comparative study. *J Strength Cond Res.* 2008;22(5):1449-57.
21. Gibson J. *The Ecological Approach to Visual Perception.* Boston: Houghton Mifflin; 1979. 332 p.
22. González-Badillo JJ, Ribas J. *Fundamentos del entrenamiento de fuerza.* Barcelona: INDE; 1999.
23. Hill-Haas SV, Coutts AJ, Rowsell GJ, Dawson BT. Generic versus small-sided game training in soccer. *Int J Sports Med.* 2009;30(9):636-42.
24. Impellizzeri FM, Rampinini E, Marcora SM. Physiological assessment of aerobic training in soccer. *J Sports Sci.* 2005;23(6):583-92.
25. Impellizzeri FM, Marcora SM, Castagna C, Reilly R, Sassi A, Iaia FM, et al. Physiological and performance effects of generic versus specific aerobic training in soccer players. *Int J Sports Med.* 2006;27:483-92.
26. Kauffman SA. *The Origins of Order: Self Organization and Selection in Evolution.* Oxford: Oxford University Press; 1993.
27. Klusemann MJ, Pyne DB, Foster C, Drinkwater EJ. Optimising technical skills and physical loading in small-sided basketball games. *J Sports Sci.* 2012;30(14):1463-71.
28. McInnes SE, Carlson JS, Jones CJ, McKenna MJ. The physiological load imposed on basketball players during competition. *J Sports Sci.* 1995;13(5):387-97.
29. Midgley AW, McNaughton LR, Wilkinson M. Is there an optimal training intensity for enhancing the maximal oxygen uptake of distance runners? Empirical research findings, current opinions, physiological rationale and practical recommendations. *Sports Med.* 2006;36(2):117-32.
30. Montgomery PG, Pyne DB, Minahan CL. The physical and physiological demands of basketball training and competition. *Int J Sports Physiol Perform.* 2010;5(1):75-86.
31. Moras G. *La preparación integral en el Voleibol.* Barcelona: Paidotribo; 1994.
32. Nassis GP, Hertzog M, Brito J. Workload assessment in soccer: An open-minded, critical thinking approach is needed. *J Strength Cond Res.* 2017;31(8):e77–e8.
33. Nicolella D, Torres-Ronda L, Saylor K, Schelling X. Validity and reliability of an accelerometer-based player tracking device. *PLoS One.* 2017;(Under Review).
34. Perše M, Kristan M, Kovačič S, Vučković G, Perš J. A trajectory-based analysis of coordinated team activity in a basketball game. *Computer Vision and Image Understanding.* 2009;113(5):612-21.

35. Rampinini E, Impellizzeri FM, Castagna C, Abt G, Chamari K, Sassi A, et al. Factors influencing physiological responses to small-sided soccer games. *J Sports Sci.* 2007;25(6):659-66.
36. Redford P. Which NBA Team Is The Tallest? : fivethirtyeight.com; 2016 [Available from: <https://fivethirtyeight.com/features/which-nba-team-is-the-tallest/>]
37. Reilly T, Morris T, Whyte G. The specificity of training prescription and physiological assessment: A review. *J Sports Sci.* 2009;27(6):575-89.
38. Remmert H. Analysis of group-tactical offensive behavior in elite basketball on the basis of a process orientated model. *Eur J Sport Sci.* 2003;3(3):1-12.
39. Sampaio J, Abrantes C, Leite N. Power, heart rate and perceived exertion to 3×3 and 4×4 basketball small-sided games. *Rev Psicol Deporte.* 2009;18:463-7.
40. Sampaio J, McGarry T, Calleja-Gonzalez J, Jimenez Saiz S, Schelling IDAX, Balciunas M. Exploring game performance in the National Basketball Association using player tracking data. *PLoS One.* 2015;10(7):e0132894.
41. Sandbakk O, Sandbakk SB, Ettema G, Welde B. Effects of intensity and duration in aerobic high-intensity interval training in highly-trained junior cross-country skiers. *J Strength Cond Res.* 2012;27(7):1974-80.
42. Scanlan AT, Wen N, Tucker PS, Borges NR, Dalbo VJ. Training mode influences the relationships between training load models during basketball conditioning. *Int J Sports Physiol Perform.* 2014;9(5):851-6.
43. Schelling X. Resistencia en baloncesto. Calidad vs Cantidad. *Minut ACEB.* 2011;9:6-8.
44. Schelling X, Torres-Ronda L. Conditioning for basketball: Quality and quantity of training. *Strength Cond J.* 2013;36(6):89-94.
45. Schelling X, Torres L. Accelerometer load profiles for basketball-specific drills in elite players. *J Sports Sci Med.* 2016;15(4):585-91.
46. Schelling X, Calleja-Gonzalez J, Torres-Ronda L, Terrados N. Using testosterone and cortisol as biomarker for training individualization in elite basketball: A 4-year follow-up study. *J Strength Cond Res.* 2015;29(2):368-78.
47. Seirul-lo F, ed. *Planificación a largo plazo en los deportes colectivos. Curso sobre Entrenamiento Deportivo en la Infancia y la Adolescencia.* 1998; Canarias, www.entrenamientodeportivo.com.
48. Stöckel T, Breslin G. The influence of visual contextual information on the emergence of the especial skill in basketball. *J Sport Exerc Psychol.* 2013;35(5):536-41.
49. Stone NM, Kilding AE. Aerobic conditioning for team sport athletes. *Sports Med.* 2009;39(8):615-42.
50. Tessitore A, Meeusen R, Cortis C, Capranica L. Effects of different recovery interventions on anaerobic performances following preseason soccer training. *J Strength Cond Res.* 2007;21(3):745-50.
51. Torres-Ronda L, Schelling X. Critical process for the implementation of technology in sport organizations. *Strength Cond J.* 2017;39(6):54-9.
52. Torres-Ronda L, Ric A, Llabres-Torres I, de Las Heras B, Schelling X. Position-dependent cardiovascular response and time-motion analysis during training drills and friendly matches in elite male basketball players. *J Strength Cond Res.* 2015;30(1):60-70.
53. Torres-Ronda L, Ric A, Llabres-Torres I, de Las Heras B, Schelling IDAX. Position-dependent cardiovascular response and time-motion analysis during training drills and friendly matches in elite male basketball players. *J Strength Cond Res.* 2016;30(1):60-70.
54. Varley MC, Aughey RJ. Acceleration profiles in elite Australian soccer. *Int J Sports Med.* 2013;34(1):34-9.
55. Weast JA, Shockley K, Riley MA. The influence of athletic experience and kinematic information on skill-relevant affordance perception. *Q J Exp Psychol.* 2011;64(4):689-706.
56. Wundersitz DW, Gastin PB, Robertson S, Davey PC, Netto KJ. Validation of a trunk-mounted accelerometer to measure peak impacts during team sport movements. *Int J Sports Med.* 2015;36(9):742-6.
57. Yarrow K, Brown P, Krakauer JW. Inside the brain of an elite athlete: The neural processes that support high achievement in sports. *Nature Rev.* 2009;10:585-96.

Chapter 24

1. Brandon R. *Investigations of the Neuromuscular Response During and Following Elite Maximum Strength and Power Type Resistance Exercise.* Scotland: University of Stirling; 2011.

2. Campbell FW, Rothwell SE, Perry MJ. Bad light stops play. *Ophthalmic and Physiological Optics*. 1987;7(2):165-7.
3. Douglas MJ, Tam N. Analysis of team performances at the ICC World Twenty20 Cup 2009. *International Journal of Performance Analysis in Sport*. 2010;10(1):47-53.
4. Genet R, Petersen C, eds. Cricket batting placement distribution analysed by bowling line and length at the 2013 ICC Champions Trophy. *World Congress of Performance Analysis of Sport X*; 2017, September 3-6: Opatija, Croatia.
5. Houghton L, Dawson B, Rubenson J. Performance in a simulated cricket batting innings (BATEX): Reliability and discrimination between playing standards. *J Sports Sci*. 2011;29(10):1097-103.
6. MacDonald DC. Performance analysis of fielding and wicket-keeping in cricket to inform strength and conditioning practice. PhD diss., Auckland University, 2015, <https://aut.researchgateway.ac.nz/bitstream/handle/10292/8819/MacDonaldD.pdf?sequence=3&isAllowed=y>.
7. McLeod P. Visual reaction time and high-speed ball games. *Perception*. 1987;16(1):49-59.
8. Mehta R. <http://www.espnricinfo.com/magazine/content/story/258645.html>: ESPNricinfo.com. 2006.
9. Najdan JM, Robins TM, Glazier SP. Determinants of success in English domestic Twenty20 cricket. *International Journal of Performance Analysis in Sport*. 2014;14(1):276-95.
10. Noakes TD, Durandt JJ. Physiological requirements of cricket. *J Sports Sci*. 2000;18(12):919-29.
11. Orchard JW, Kountouris A, Sims K. Incidence and prevalence of elite male cricket injuries using updated consensus definitions. *Open Access Journal of Sports Medicine*. 2016;7:187.
12. Orchard JW, James T, Portus M, Kountouris A, Dennis R. Fast bowlers in cricket demonstrate up to 3- to 4-week delay between high workloads and increased risk of injury. *Amer J Sports Med*. 2009;37(6):1186-92.
13. Petersen C, Genet R, eds. Captain and coaches playbook app: A practical display of fielding data. *5th World Congress of Science and Medicine in Cricket*; 2015, March 23-27: Sydney, Australia.
14. Petersen C, Pyne DB, Portus MJ, Dawson B. Analysis of Twenty/20 cricket performance during the 2008 Indian Premier League. *International Journal of Performance Analysis in Sport*. 2008;8(3):63-9.
15. Petersen C, Pyne DB, Portus MR, Dawson B. Quantifying positional movement patterns in Twenty20 cricket. *International Journal of Performance Analysis in Sport*. 2009;9(2):165-70.
16. Petersen CJ. Comparison of performance at the 2007 and 2015 Cricket World Cups. *Int J Sports Sci Coach*. 2017;12(3):404-10.
17. Petersen CJ, Pyne D, Dawson B, Portus M, Kellett A. Movement patterns in cricket vary by both position and game format. *J Sports Sci*. 2010;28(1):45-52.
18. Petersen CJ, Pyne DB, Dawson BT, Kellett AD, Portus MR. Comparison of training and game demands of national level cricketers. *J Strength Cond Res*. 2011;25(5):1306-11.
19. Petersen CJ, Portus MR, Pyne DB, Dawson BT, Cramer MN, Kellett AD. Partial heat acclimation in cricketers using a 4-day high intensity cycling protocol. *Int J Sports Physiol Perform*. 2010;5(4):535-45.
20. Portus M, Kellett A, Karppinen S, Timms S. Physiological tests for elite athletes, 2nd Edition. In: Tanner R, Gore C, eds. *Chapter 18 Cricket Players*. Champaign, IL: Human Kinetics; 2012: 289-97.
21. Pote L, Christie CJ. Physiological responses of batsmen during a simulated One Day International century. *South African Journal of Sports Medicine*. 2016;28(2):39-42.
22. Pyke F, Davis K. *Cutting Edge Cricket*. Champaign, IL: Human Kinetics; 2010.
23. Renshaw I, Fairweather MM. Cricket bowling deliveries and the discrimination ability of professional and amateur batters. *J Sports Sci*. 2000;18(12):951-7.
24. Tyndall N. Personal communication. 2017.
25. Vickery W, Dascombe B, Duffield R, Kellett A, Portus M. The influence of field size, player number and rule changes on the physiological responses and movement demands of small-sided games for cricket training. *J Sports Sci*. 2013;31(6):629-38.

26. Vickery W, Duffield R, Crowther R, Beakley D, Blanch P, Dascombe BJ. Comparison of the physical and technical demands of cricket players during training and match-play. *J Strength Cond Res.* 2018;32(3):821-29.
27. Wigmore T. Northants' T20 nous a tribute to moneyball approach. ESPNcricinfo.com. 2016. Available from: <http://www.espncricinfo.com/county-cricket-2017/content/story/1070189.html>.
28. Woolmer B, Noakes T, Moffett H. *Bob Woolmer's Art and Science of Cricket*. London: New Holland Publishers; 2008.
9. Spencer M, Bishop D, Lawrence S. Longitudinal assessment of the effects of field-hockey training on repeated sprint ability. *J Sci Med Sport.* 2004;7(3):323-34.
10. Spencer M, Lawrence S, Rechichi C, Bishop D, Dawson B, Goodman C. Time-motion analysis of elite field hockey, with special reference to repeated-sprint activity. *J Sports Sci.* 2004;22(9):843-50.
11. Sweet TW, Foster C, McGuigan MR, Brice G. Quantitation of resistance training using the session rating of perceived exertion method. *J Strength Cond Res.* 2004;18(4):796-802.

Chapter 25

1. Abbott HA. Positional and Match Action Profiles of Elite Women's Field Hockey Players in Relationship to the 2015 FIH Rule Changes. Electronic Theses and Dissertations. Paper 3092: University of Tennessee; 2016.
2. Buchheit M. The 30-15 Intermittent Fitness Test: 10 year review. *Myorobie Journal.* 2010;1(September):1-9.
3. Cormack SJ, Mooney MG, Morgan W, McGuigan MR. Influence of neuromuscular fatigue on accelerometer load in elite Australian football players. *Int J Sports Physiol Perform.* 2013;8(4):373-8.
4. Impellizzeri FM, Rampinini E, Coutts AJ, Sassi A, Marcora SM. Use of RPE-based training load in soccer. *Med Sci Sports Exerc.* 2004;36(6):1042-7.
5. Jennings D, Cormack SJ, Coutts AJ, Aughey RJ. GPS analysis of an international field hockey tournament. *Int J Sports Physiol Perform.* 2012;7(3):224-31.
6. Jones B, Hamilton DK, Cooper CE. Muscle oxygen changes following sprint interval cycling training in elite field hockey players. *PLoS One.* 2015;10(3):e0120338.
7. Lemmink KA, Visscher SH. Role of energy systems in two intermittent field tests in women field hockey players. *J Strength Cond Res.* 2006;20(3):682-8.
8. McGuigan MR, Doyle TL, Newton M, Edwards DJ, Nimphius S, Newton RU. Eccentric utilization ratio: Effect of sport and phase of training. *J Strength Cond Res.* 2006;20(4):992-5.

Chapter 26

1. Bell G, Game A, Bouchard J, Reid C, Gervais P, Snyder Miller G. Near maximal heart rate responses during a varsity ice hockey game. *Applied Physiology, Nutrition and Metabolism.* 2011;36(S2).
2. Besson C, Buchheit M, Praz M, Deriaz O, Millet GP. Cardiorespiratory responses to the 30-15 intermittent ice test. *Int J Sports Physiol Perform.* 2013;8(2):173-80.
3. Bracko MR, Fellingham GW, Hall LT, Fisher AG, Cryer W. Performance skating characteristics of professional ice hockey forwards. *Sports Medicine, Training and Rehabilitation.* 1998;8(3):251-63.
4. Buchheit M, Lefebvre B, Laursen PB, Ahmaidi S. Reliability, usefulness, and validity of the 30-15 Intermittent Ice Test in young elite ice hockey players. *J Strength Cond Res.* 2011;25(5):1457-64.
5. Cox MH, Miles DS, Verde TJ, Rhodes EC. Applied physiology of ice hockey. *Sports Med.* 1995;19(3):184-201.
6. Girard O, Mendez-Villanueva A, Bishop D. Repeated-sprint ability—Part I: Factors contributing to fatigue. *Sports Med.* 2011;41(8):673-94.
7. Jackson JK. *Time-Motion Characteristics and Heart Rate Profiles Displayed by Female University Ice Hockey Players*. Edmonton, Canada: University of Alberta; 2015.
8. Janot JM, Beltz NM, Dalleck LD. Multiple off-ice performance variables predict on-ice skating performance in male and female Division III ice hockey players. *J Sports Sci Med.* 2015;14(3):522-9.

9. Manners TW. Sport-specific training for ice hockey. *Strength Cond J*. 2004;April.
 10. Montgomery DL. Physiology of ice hockey. *Sports Med*. 1988;5(2):99-126.
 11. Peterson BJ, Fitzgerald JS, Dietz CC, Ziegler KS, Baker SE, Snyder EM. Off-ice anaerobic power does not predict on-ice repeated shift performance in hockey. *J Strength Cond Res*. 2016;30(9):2375-81.
 12. Quinney HA, Dewart R, Game A, Snydmiller G, Warburton D, Bell G. A 26 year physiological description of a National Hockey League team. *Appl Physiol Nutr Metab*. 2008;33(4):753-60.
- ### Chapter 27
1. Al Haddad H, Laursen PB, Chollet D, Ahmaidi S, Buchheit M. Reliability of resting and postexercise heart rate measures. *Int J Sports Med*. 2011;32(8):598-605.
 2. Bompa TO, Carrera M. *Periodization Training for Sports*, 2nd Edition. Champaign, IL: Human Kinetics; 2005.
 3. Buchheit M. Individualizing high-intensity interval training in intermittent sport athletes with the 30-15 Intermittent Fitness Test. *NSCA Hot Topic Series www.nsca-liftorg*. 2011; November.
 4. Buchheit M. Should we be recommending repeated sprints to improve repeated-sprint performance? *Sports Med*. 2012;42(2):169-72.
 5. Buchheit M. Programming high-intensity training in handball. *Aspetar-Sports Medicine Journal*. 2014;3(Targeted topic-sports medicine in handball):120-8.
 6. Buchheit M. Monitoring training status with HR measures: Do all roads lead to Rome? *Front Physiol*. 2014;27(5):73.
 7. Buchheit M, Laursen PB. High-intensity interval training, solutions to the programming puzzle: Part I: Cardiopulmonary emphasis. *Sports Med*. 2013;43(5):313-38.
 8. Buchheit M, Laursen PB. High-intensity interval training, solutions to the programming puzzle. Part II: Anaerobic energy, neuromuscular load and practical applications. *Sports Med*. 2013;43(10):927-54.
 9. Buchheit M, Lepretre PM, Behaegel AL, Millet GP, Ahmaidi S. Cardiorespiratory responses during running and sport-specific exercises in handball players. *J Sci Med Sport* 2008; 12(3):399-405.
 10. Buchheit M, Mendez-Villanueva A, Quod MJ, Quesnel T, Ahmaidi S. Improving acceleration and repeated sprint ability in well-trained adolescent handball players: Speed vs. sprint interval training. *Int J Sports Physiol Perform*. 2010;5(2):152-64.
 11. Buchheit M, Morgan W, Wallace J, Bode M, Poulos N. Physiological, psychometric, and performance effects of the Christmas break in Australian football. *Int J Sports Physiol Perform*. 2015;10(1):120-3.
 12. Buchheit M, Millet GP, Parisy A, Pourchez S, Laursen PB, Ahmaidi S. Supramaximal training and post-exercise parasympathetic reactivation in adolescents. *Med Sci Sports Exerc*. 2008;40(2):362-71.
 13. Buchheit M, Laursen PB, Kuhnle J, Ruch D, Renaud C, Ahmaidi S. Game-based training in young elite handball players. *Int J Sports Med*. 2009;30(4):251-8.
 14. Buchheit M, Lepretre PM, Behaegel AL, Millet GP, Cuvelier G, Ahmaidi S. Cardiorespiratory responses during running and sport-specific exercises in handball players. *J Sci Med Sport*. 2009;12(3):399-405.
 15. Buchheit M, Al Haddad H, Lepretre PM, Millet G, Newton M, Ahmaidi S. Cardiorespiratory and cardiac autonomic responses to 30-15: Intermittent fitness test. *J Strength Cond Res*. 2009;23(1):93-100.
 16. Cardinale M, Whiteley R, Hosny AA, Popovic N. Activity profiles and positional differences of handball players during the World Championships in Qatar 2015. *Int J Sports Physiol Perform*. 2017;12(7):908-15.
 17. Corvino M, Tessitore A, Minganti C, Šibila M. Effect of court dimensions on players' external and internal load during small-sided handball games *J Sports Sci & Med*. 2014;13(2):297-303.
 18. Delgado-Bordonau JL, Mendez-Villanueva A. The Tactical Periodization Model. In: Sum ME, ed. *Fitness in Soccer: The Science and Practical Application*. Groningen, The Netherlands: Moveo Ergo Sum; 2014.
 19. Dello Iacono A, Martone D, Zagatto AM, Meckel Y, Sindiani M, Milic M, et al. Effect of contact and no-contact small-sided games on elite handball players. *J Sports Sci*. 2016;35(1):1-9.
 20. Francis C. *Training for Speed*. Bellingham, WA: Faccioni Speed & Conditioning Consultants; 1997.

21. Gorostiaga EM, Granados C, Ibanez J, Izquierdo M. Differences in physical fitness and throwing velocity among elite and amateur male handball players. *Int J Sports Med*. 2005;26(3):225-32.
 22. Hopkins WG, Marshall SW, Batterham AM, Hanin J. Progressive statistics for studies in sports medicine and exercise science. *Med Sci Sports Exerc*. 2009;41(1):3-13.
 23. Impellizzeri FM, Rampinini E, Coutts AJ, Sassi A, Marcora SM. Use of RPE-based training load in soccer. *Med Sci Sports Exerc*. 2004;36(6):1042-7.
 24. Karcher C, Buchheit M. On-court demands of elite handball, with special reference to playing positions. *Sports Med*. 2014;44(6):797-814.
 25. Loftin M, Anderson P, Lytton L, Pittman P, Warren B. Heart rate response during handball singles match-play and selected physical fitness components of experienced male handball players. *J Sports Med Phys Fitness*. 1996;36(2):95-9.
 26. Luteberget LS, Spencer M. High-intensity events in international women's team handball matches. *Int J Sports Physiol Perform*. 2017;12(1):56-61.
 27. Rannou F, Prioux J, Zouhal H, Gratas-Delamarche A, Delamarche P. Physiological profile of handball players. *J Sports Med Phys Fitness*. 2001;41(3):349-53.
 28. Saw AE, Main LC, Gastin PB. Monitoring the athlete training response: Subjective self-reported measures trump commonly used objective measures: A systematic review. *Br J Sports Med*. 2016;50(5):281-91.
 29. Spencer M, Bishop D, Dawson B, Goodman C. Physiological and metabolic responses of repeated-sprint activities: Specific to field-based team sports. *Sports Med*. 2005;35(12):1025-44.
 30. Stanley J, Peake JM, Buchheit M. Cardiac parasympathetic reactivation following exercise: Implications for training prescription. *Sports Med*. 2013;43(12):1259-77.
 31. Ufland P, Ahmaidi S, Buchheit M. Repeated-sprint performance, locomotor profile and muscle oxygen uptake recovery. *Int J Sport Med*. 2013;34(10):924-30.
 32. Wik EH, Luteberget LS, Spencer M. Activity profiles in international female team handball using PlayerLoad. *Int J Sports Physiol Perform*. 2017;12(7):934-42.
- ## Chapter 28
1. Austin D, Gabbett T, Jenkins D. The physical demands of Super 14 Rugby Union. *J Sci Med Sport*. 2011;14(3):259-63.
 2. Bishop D, Girard O, Mendez-Villanueva A. Repeated-sprint ability—Part II: Recommendations for training. *Sports Med*. 2011;41(9):741-56.
 3. Clemente FM, Wong del P, Martins FM, Mendes RS. Acute effects of the number of players and scoring method on physiological, physical, and technical performance in small-sided soccer games. *Res Sports Med*. 2014;22(4):380-97.
 4. Cunniffe B, Proctor W, Baker JS, Davies B. An evaluation of the physiological demands of elite Rugby Union using global positioning system tracking software. *J Strength Cond Res*. 2009;23(4):1195-203.
 5. Cunningham DJ, Shearer DA, Drawer S, Pollard B, Eager R, Taylor N, et al. Movement demands of elite under-20s and senior international Rugby Union players. *PLoS One*. 2016;11(11):e0164990.
 6. Deutsch MU, Kearney GA, Rehrer NJ. Time-motion analysis of professional Rugby Union players during match-play. *J Sports Sci*. 2007;25(4):461-72.
 7. Duthie G, Pyne D, Hooper S. Time-motion analysis of 2001 and 2002 Super 12 rugby. *J Sports Sci*. 2005;23(5):523-30.
 8. Duthie GM, Pyne DB, Hooper SL. Applied physiology and game analysis of Rugby Union. *Sports Med*. 2003;33(13):973-91.
 9. Eaves S, Hughes M. Patterns of play of international Rugby Union teams before and after the introduction of professional status. *International Journal of Performance Analysis in Sport*. 2003;3(2):103-11.
 10. Eaves SJ, Hughes MD, Lamb KL. The consequences of the introduction of professional playing status on game action variables in international Northern Hemisphere Rugby Union football. *International Journal of Performance Analysis in Sport*. 2005;5(2):58-86.
 11. Fuller CW, Brooks JH, Cancea RJ, Hall J, Kemp SP. Contact events in Rugby Union and their propensity to cause injury. *Br J Sports Med*. 2007;41(12):862-7.
 12. Gabbett TJ, Jenkins D, Abernethy B. Game-based training for improving skill and physical fitness in team sport athletes. *Int J Sports Sci Coach*. 2009;4(2):273-83.

13. Hogarth LW, Burkett BJ, McKean MR. Match demands of professional rugby football codes: A review from 2008 to 2015. *Int J Sports Sci Coach*. 2016;11(3):451-63.
14. James N, Mellalieu SD, Jones NM. The development of position-specific performance indicators in professional Rugby Union. *J Sports Sci*. 2005;23(1):63-72.
15. Jones MR, West DJ, Crewther BT, Cook CJ, Kilduff LP. Quantifying positional and temporal movement patterns in professional Rugby Union using global positioning system. *Eur J Sport Sci*. 2015;15(6):488-96.
16. Jones NMP, Mellalieu SD, James N. Team performance indicators as a function of winning and losing in Rugby Union. *International Journal of Performance Analysis in Sport*. 2004;4(1):61-71.
17. Jones RM, Cook CC, Kilduff LP, Milanovic Z, James N, Sporis G, et al. Relationship between repeated sprint ability and aerobic capacity in professional soccer players. *Scientific World Journal*. 2013;2013:952350.
18. Owen AL, Wong DP, Paul D, Dellal A. Physical and technical comparisons between various-sided games within professional soccer. *Int J Sports Med*. 2014;35(4):286-92.
19. Quarrie K, Handcock P, Waller AE, Chalmers DJ, Toomey MJ, Wilson BD. The New Zealand rugby injury and performance project. III. Anthropometric and physical performance characteristics of players. *Br J Sports Med*. 1995;29(4):263-70.
20. Quarrie KL, Hopkins WG. Changes in player characteristics and match activities in Bledisloe Cup Rugby Union from 1972 to 2004. *J Sports Sci*. 2007;25(8):895-903.
21. Quarrie KL, Hopkins WG, Anthony MJ, Gill ND. Positional demands of international Rugby Union: Evaluation of player actions and movements. *J Sci Med Sport*. 2013;16(4):353-9.
22. Roberts SP, Trewartha G, Higgitt RJ, El-Abd J, Stokes KA. The physical demands of elite English Rugby Union. *J Sports Sci*. 2008;26(8):825-33.
23. Smart D, Hopkins WG, Quarrie KL, Gill N. The relationship between physical fitness and game behaviours in Rugby Union players. *Eur J Sport Sci*. 2014;14(Suppl 1):S8-S17.
24. Smith TB, Hébert-Losier K, McClymont D. An examination of the jump-and-lift factors influencing the time to reach peak catch height during a Rugby Union lineout. *J Sports Sci*. 2017;1-7.
25. Swaby R, Jones PA, Comfort P. Relationship between maximum aerobic speed performance and distance covered in Rugby Union games. *J Strength Cond Res*. 2016;30(10):2788-93.
26. Tee JC, Lambert MI, Coopoo Y. GPS comparison of training activities and game demands of professional Rugby Union. *Int J Sports Sci Coach*. 2016;11(2):200-11.
27. Tomlin DL, Wenger HA. The relationship between aerobic fitness and recovery from high intensity intermittent exercise. *Sports Med*. 2001;31(1):1-11.
28. Twist C, Worsfold P. *The Science of Rugby*. London: Taylor & Francis; 2015.

Chapter 29

1. Bellenger C. Predicting maximal aerobic speed through set distance time-trials. *Eur J Appl Physiol*. 2015;115(12):2593-8.
2. Bompa TO, Carrera M. *Periodization Training for Sports*, 2nd Edition. Champaign, IL: Human Kinetics; 2005.
3. Buchheit M. Individualizing high-intensity interval training in intermittent sport athletes with the 30-15 Intermittent Fitness Test. *NSCA Hot Topic Series wwwnsca-liftorg*. 2011;November.
4. Buchheit M. Monitoring training status with HR measures: Do all roads lead to Rome? *Front Physiol*. 2014;27(5):73.
5. Buchheit M, Laursen PB. High-intensity interval training, solutions to the programming puzzle. Part II: Anaerobic energy, neuromuscular load and practical applications. *Sports Med*. 2013;43(10):927-54.
6. Buchheit M, Laursen PB. High-intensity interval training, solutions to the programming puzzle: Part I: Cardiopulmonary emphasis. *Sports Med*. 2013;43(5):313-38.
7. Buchheit M, Simpson B. Player tracking technology: Half-full or half-empty glass? *Int J Sports Physiol Perform*. 2017;12(Suppl 2):S235-S41.
8. Buchheit M, Mendez-Villanueva A, Quod MJ, Quesnel T, Ahmaidi S. Improving acceleration and repeated sprint ability in well-trained adolescent handball players: speed vs. sprint interval training. *Int J Sports Physiol Perform*. 2010;5(2):152-64.

9. Buchheit M, Morgan W, Wallace J, Bode M, Poulos N. Physiological, psychometric, and performance effects of the Christmas break in Australian Football. *Int J Sports Physiol Perform*. 2015;10(1):120-3.
10. Clarke AC, Anson J, Pyne D. Physiologically based GPS speed zones for evaluating running demands in Women's Rugby Sevens. *J Sports Sci*. 2015;33(11):1101-8.
11. Coughlan GF. Physical game demands in elite Rugby Union: A global positioning system analysis and possible implications for rehabilitation. *Journal of Orthopaedic and Sports Physical Therapy*. 2011;41(8):600-5.
12. Duthie GM. A framework for the physical development of elite Rugby Union players. *Int J Sports Physiol Perform*. 2006;1(1):2-13.
13. Francis C. *Training for Speed*. Bellingham, WA: Faccioni Speed & Conditioning Consultants; 1997.
14. Gabbett TJ. The training-injury prevention paradox: Should athletes be training smarter and harder? *Br J Sports Med*. 2016;50(5):273-80.
15. Higham DG, Pyne DB, Anson JM, Eddy A. Movement patterns in Rugby Sevens: Effects of tournament level, fatigue and substitute players. *J Sci Med Sport*. 2012;15(3):277-82.
16. Higham DG, Pyne DB, Anson JM, Eddy A. Physiological, anthropometric, and performance characteristics of Rugby Sevens players. *Int J Sports Physiol Perform*. 2013;8(1):19-27.
17. Higham DG, Pyne DB, Anson JM, Hopkins WG, Eddy A. Comparison of activity profiles and physiological demands between international Rugby Sevens matches and training. *J Strength Cond Res*. 2016;30(5):1287-94.
18. McLellan CP, Lovell DI. Performance analysis of professional, semiprofessional, and junior elite rugby league match-play using global positioning systems. *J Strength Cond Res*. 2013;27(12):3266-74.
19. Mitchell JA. Variable changes in body composition, strength and lower body power during an international sevens season *J Strength Cond Res*. 2016;30(4):1127-36.
20. Murray AM, Varley MC. Activity profile of International Rugby Sevens: Effect of score line, opponent, and substitutes. *Int J Sports Physiol Perform*. 2015;10(6):791-801.
21. Rienzi E. Investigation of anthropometric and work-rate profiles of Rugby Sevens players. *Journal of Sports Medicine and Physical Fitness*. 1999;39(2):160-4.
22. Robineau J, Lacombe M, Piscione J, Bigard X, Babault N. Concurrent training in Rugby Sevens: Effects of high-intensity interval exercises. *Int J Sports Physiol Perform*. 2017;12(3):336-44.
23. Ross A, Gill N, Cronin J. Match analysis and player characteristics in Rugby Sevens. *Sports Med (Auckland, NZ)*. 2014;44(3):357-67.
24. Ross A, Gill N, Cronin J. The match demands of international rugby sevens. *J Sports Sci*. 2015;33(10):1035-41.
25. Ross A, Gill ND, Cronin JB. A comparison of the match demands of international and provincial Rugby Sevens. *Int J Sports Physiol Perform*. 2015;10(6):786-90.
26. Ross A, Gill N, Cronin J, Malcata R. The relationship between physical characteristics and match performance in Rugby Sevens. *Eur J Sport Sci*. 2015;15(6):565-71.
27. Saw AE, Main LC, Gastin PB. Monitoring the athlete training response: Subjective self-reported measures trump commonly used objective measures: A systematic review. *Br J Sports Med*. 2016;50(5):281-91.
28. Schuster J, Howells D, Robineau J, Couderc A, Natera A, Lumley N, et al. Physical preparation recommendations for elite Rugby Sevens performance. *Int J Sports Physiol Perform*. 2017:1-42.
29. Suarez-Arrones L, Arenas C, López G, Requena B, Terrill O, Mendez-Villanueva A. Positional differences in match running performance and physical collisions in men Rugby Sevens. *Int J Sports Physiol Perform*. 2014;9(2):316-23.
30. Suarez-Arrones L, Núñez J, Sáez de Villareal E, Gálvez J, Suarez-Sanchez G, Munguía-Izquierdo D. Repeated-high-intensity-running activity and internal training load of elite Rugby Sevens players during international matches: A comparison between halves. *Int J Sports Physiol Perform*. 2016;11(4):495-9.
31. Suarez-Arrones LJ. Running demands and heart rate responses in men Rugby Sevens. *J Strength Cond Res*. 2012;26(11):3155-9.
32. Waldron M, Twist C, Highton J, Worsfold P, Daniels M. Movement and physiological match demands of elite rugby league using portable

global positioning systems. *J Sports Sci.* 2011;29(11):1223-30.

Chapter 30

1. Akenhead R, Nassis GP. Training load and player monitoring in high-level football: Current practice and perceptions. *Int J Sports Physiol Perform.* 2016;11(5):587-93.
2. Barnes C, Archer DT, Hogg B, Bush M, Bradley PS. The evolution of physical and technical performance parameters in the English Premier League. *Int J Sports Med.* 2014;35(13):1095-100.
3. Buchheit M. Individualizing high-intensity interval training in intermittent sport athletes with the 30-15 Intermittent Fitness Test. *NSCA Hot Topic Series wwwnsca-liftorg.* 2011;November.
4. Buchheit M. Monitoring training status with HR measures: Do all roads lead to Rome? *Front Physiol.* 2014;27(5):73.
5. Buchheit M, Laursen PB. High-intensity interval training, solutions to the programming puzzle: Part I: Cardiopulmonary emphasis. *Sports Med.* 2013;43(5):313-38.
6. Buchheit M, Laursen PB. High-intensity interval training, solutions to the programming puzzle. Part II: Anaerobic energy, neuromuscular load and practical applications. *Sports Med.* 2013;43(10):927-54.
7. Buchheit M, Simpson BM. Player tracking technology: Half-full or half-empty glass? *Int J Sports Physiol Perform.* 2017;12(Suppl 2):S235–S41.
8. Buchheit M, Haydar B, Ahmaidi S. Repeated sprints with directional changes: Do angles matter? *J Sports Sci.* 2012;30(6):555-62.
9. Buchheit M, Cholley Y, Lambert P. Psychometric and physiological responses to a preseason competitive camp in the heat with a 6-hour time difference in elite soccer players. *Int J Sports Physiol Perform.* 2016;11(2):176-81.
10. Buchheit M, Mendez-Villanueva A, Simpson BM, Bourdon PC. Match running performance and fitness in youth soccer. *Int J Sports Med.* 2010;31(11):818-25.
11. Buchheit M, Lacome M, Cholley Y, Simpson BM. Neuromuscular responses to conditioned soccer sessions assessed via GPS-embedded accelerometers: Insights into tactical periodization. *Int J Sports Physiol Perform.* Sept 2017;1-21.
12. Buchheit M, Morgan W, Wallace J, Bode M, Poulos N. Physiological, psychometric, and performance effects of the Christmas break in Australian football. *Int J Sports Physiol Perform.* 2015;10(1):120-3.
13. Carling C. Interpreting physical performance in professional soccer match-play: Should we be more pragmatic in our approach? *Sports Med.* 2013;43(8):655-63.
14. Carling C, Bloomfield J. The effect of an early dismissal on player work-rate in a professional soccer match. *J Sci Med Sport.* 2010;13(1):126-8.
15. Delgado-Bordonau JL, Mendez-Villanueva A. Tactical Periodization: Mourinho's best-kept secret? *Soccer Journal.* May/June 2015. <http://growingthegame.ca/wp/wp-content/uploads/2015/07/Tactical-Periodization.pdf>.
16. Dellal A, Lago-Penas C, Wong del P, Chamari K. Effect of the number of ball contacts within bouts of 4 vs. 4 small-sided soccer games. *Int J Sports Physiol Perform.* 2011;6(3):322-33.
17. Dellal A, Varliette C, Owen A, Chirico EN, Pialoux V. Small-sided games versus interval training in amateur soccer players: Effects on the aerobic capacity and the ability to perform intermittent exercises with changes of direction. *J Strength Cond Res.* 2012;26(10):2712-20.
18. Dellal A, Chamari K, Pintus A, Girard O, Cotte T, Keller D. Heart rate responses during small-sided games and short intermittent running training in elite soccer players: A comparative study. *J Strength Cond Res.* 2008;22(5):1449-57.
19. Fyfe JJ, Bishop DJ, Stepto NK. Interference between concurrent resistance and endurance exercise: molecular bases and the role of individual training variables. *Sports Med.* 2014;44(6):743-62.
20. Gabbett TJ. The training-injury prevention paradox: Should athletes be training smarter and harder? *Br J Sports Med.* 2016;50(5):273-80.
21. Hader K, Mendez-Villanueva A, Williams B, Ahmaidi S, Buchheit M. Changes of direction during high-intensity intermittent runs: Neuromuscular and metabolic responses. *BMC Sports Sci Med Rehabil.* 2014;6(1):2.
22. Hader K, Mendez-Villanueva A, Palazzi D, Ahmaidi S, Buchheit M. Metabolic power requirement of change of direction speed in

- young soccer players: Not all is what it seems. *PLoS One*. 2016;11(3):e0149839.
23. Hill-Haas SV, Coutts AJ, Rowsell GJ, Dawson BT. Generic versus small-sided game training in soccer. *Int J Sports Med*. 2009;30(9):636-42.
 24. Hill-Haas SV, Dawson B, Impellizzeri FM, Coutts AJ. Physiology of small-sided games training in football: A systematic review. *Sports Med*. 2011;41(3):199-220.
 25. Impellizzeri FM, Rampinini E, Coutts AJ, Sassi A, Marcora SM. Use of RPE-based training load in soccer. *Med Sci Sports Exerc*. 2004;36(6):1042-7.
 26. Impellizzeri FM, Marcora SM, Castagna C, Reilly T, Sassi A, Iaia FM, et al. Physiological and performance effects of generic versus specific aerobic training in soccer players. *Int J Sports Med*. 2006;27(6):483-92.
 27. Lacome M, Simpson BM, Buchheit M. Locomotor and heart rate responses of floaters during small-sided games in elite soccer players: Effect of pitch size and goal keepers. *Int J Sports Physiol Perform*. Sept 2017;1-13.
 28. Lacome M, Simpson BM, Cholley Y, Lambert P, Buchheit M. Small sided games in elite soccer: Does one size fit all? *Int J Sports Physiol Perform*. Jul 2017;1-24.
 29. Mayer N, Bosquet L, Plaine F, Marles A, Jullien H, Lambert P, et al., eds. Reproducibility of physiological, neuromuscular and perceptual responses to small-sided games in highly-trained young soccer players. European College of Sport Science; 2013; Barcelona, June 26-29.
 30. Mendez-Villanueva A, Buchheit M, Simpson BM, Bourdon PC. Match play intensity distribution in youth soccer. *Int J Sport Med*. 2013;34(2):101-10.
 31. Mendez-Villanueva A, Buchheit M, Simpson B, Peltola E, Bourdon P. Does on-field sprinting performance in young soccer players depend on how fast they can run or how fast they do run? *J Strength Cond Res*. 2011;25(9):2634-8.
 32. Morgans R, Di Michele R, Drust B. Soccer match-play represents an important component of the power training stimulus in premier league players. *Int J Sports Physiol Perform*. 2018;13(5):665-7.
 33. Rabbani A, Buchheit M. Heart rate-based versus speed-based high-intensity interval training in young soccer players. In Favero T, Drust B, Dawson B, eds. *International Research in Science and Soccer II*. Abingdon, UK: Routledge; 2015.
 34. Rampinini E, Impellizzeri F, Castagna C, Abt G, Chamari K, Sassi A, et al. Factors influencing physiological responses to small-sided soccer games. *J Sports Sci*. 2007;6(25):659-66.
 35. Rebelo AN, Silva P, Rago V, Barreira D, Krustup P. Differences in strength and speed demands between 4v4 and 8v8 small-sided football games. *J Sports Sci*. 2016;34(24): 2246-54.
 36. Reilly T, Bangsbo J, Franks A. Anthropometric and physiological predispositions for elite soccer. *J Sports Sci*. 2000;18(9):669-83.
 37. Saw AE, Main LC, Gastin PB. Monitoring the athlete training response: Subjective self-reported measures trump commonly used objective measures: A systematic review. *Br J Sports Med*. 2016;50(5):281-91.
 38. Spencer M, Bishop D, Dawson B, Goodman C. Physiological and metabolic responses of repeated-sprint activities: Specific to field-based team sports. *Sports Med*. 2005;35(12):1025-44.
 39. Ufland P, Ahmaidi S, Buchheit M. Repeated-sprint performance, locomotor profile and muscle oxygen uptake recovery. *Int J Sport Med*. 2013;34(10):924-30.

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